MAT367 Lecture Notes

Arky!! :3c

'26 Winter Semester

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§1 Day 1: Recap of Preliminaries (Jan. 6, 2026)

Today's class can be followed more precisely on §1.2 to §1.4 of our textbook by Gross and Meinrenken. The slogan of this class is that a manifold is something that locally looks like $\mathbb{R}^n$. Specifically, an n-manifold can be covered n-dimensional charts $(U \subset M) \to \mathbb{R}^n$, with our main motivating example being solutions sets to equations. Recall the implicit function theorem,

Theorem 1.1. Given a smooth function $f : \mathbb{R}^{n+1} \to \mathbb{R}$, consider the solution set $f(x_1, \ldots, x_{n+1}) = 0$ and a point $p \in \mathbb{R}^n$ such that $\nabla f(p) \neq 0$; then, for $(x_1, \ldots, x_{n+1})$ in said solution set near p, we can represent solutions as $(x_1, \ldots, x_n, g(x_1, \ldots, x_n))$, where g is also a smooth function.

In particular, if 0 is a regular value[1] of f, then we can cover $\{x \mid f(x) = 0\}$ by graphs/charts. We present some examples;

(i) Let $f(x, y) = xy$; then $\ker f$ is precisely the x and y axes, which is not a manifold, because it does not look like $\mathbb{R}^n$ (for any n) near the origin.

(ii) Let $f(x, y) = y - x^{2/3}$; then $\ker f$ can be graphed in desmos as $y = x^{2/3}$, which is not a smooth manifold because of its behavior at 0.

(iii) The n-sphere $S^n = \{x \in \mathbb{R}^n \mid x_1^2 + \cdots + x_{n+1}^2 = 1\}$ can be regarded as the level set of the ℓ^2-norm, for which $S^0 = \{\pm 1\} \subset \mathbb{R}$, S^1 is a circle, S^2 is the usual sphere. Note that we may use the stereographic projection as seen in complex analysis, to view S^3 (and any of the previous or subsequence S^n) as $\mathbb{R}^3 \cup \{\infty\}$.

(iv) The 2-dimensional torus T^2 is the surface of revolution obtained from a circle of radius r and R about an axis of revolution. It can be regarded as a level set by writing
$$T^2 = \{(x, y, z) \in \mathbb{R}^3 \mid (\sqrt{x^2 + y^2} - R)^2 + z^2 = r\}.$$

(v) The Möbius strip can't be a part of a level set (at a regular value) because level sets are orientable (2-sided), while the strip is not.

(vi) The Klein bottle is also not orientable; it is closed (doesn't have a boundary), and doesn't embed into $\mathbb{R}^3$. It can be immersed into $\mathbb{R}^3$, i.e., locally embedded but not globally, as seen in the textbook.

Theorem 1.2 (Whitney Embedding Theorem)**.** Every n-manifold has an embedding in $\mathbb{R}^{2n}$.

In this class, we prefer to deal with intrinsic descriptions of manifolds rather than extrinsic ones; a good motivation is given on p.7 in the textbook with respect to our 2-torus.

Now, consider M to be the rotations of a ball. We call this a configuration spcae; i.e., its points are a way of configuring another object. How do we put coordinates on a piece of this space? To start, we wish to describe M; one possible way is by considering the 3×3 orthonormal matrices, i.e., $\mathrm{SO}(3)$; another way is to first designate a point on the unit sphere as the north pole N (of which there are two degrees of freedom in this choice), then choosing where the vector $(1, 0, 0)$ at N is mapped to (of which we have one degree of freedom).

In this manner, we may regard M as a 3-dimensional manifold and define coordinates for

[1] note to self: what's a regular value?

points near the identity.

A second example is given by considering linkages, which are collections of line segments and joints (p.8-10); suppose we have four segments given by $n_1, \ldots, n_4$, to be regarded as vectors in $\mathbb{R}^2$; without additional constraints imposed, we see that this can be regarded as 8-dimensional. In imposing the conditions on *making* $n_1, \ldots, n_4$ a linkage, we see that $\|n_i\|$ for each $i = 1, \ldots, 4$ is fixed (whence 4 less dimensions), $n_1 + n_2 + n_3 + n_4 = 0$ (whence one less), and n_1 is fixed (whence one less again), so such a linkage can be regarded as a 2 dimensional manifold with coordinates θ and φ.

Finally, denote $\mathbb{R}\mathrm{P}^2$ the real projective plane, given by lines passing through the origin in $\mathbb{R}^3$; equivalently, we may regard this as the pairs of antipodal $\{x, -x\}$ points in S^2.

Fact 1.3. "Closed" (equivalently, compact) surfaces are easy to enumerate.

Fedya then said something about orientability, but that's left in the textbook and easier to read there.

We now wish to define manifolds. For a first attempt, we want to say that a smooth manifold is a set M covered by a set of smooth charts $\varphi_i : (U_i \subset M) \to \mathbb{R}^n$, such that each $p \in M$ is covered by some U_i. However, we run into a problem; what does "smooth" mean? $f : U \to \mathbb{R}^n$ recall that a function is said to be smooth if it has partial derivatives of all orders. A function $f : U \to V$ is a diffeomorphism if it's a smooth bijectino and its inverse is also smooth.

Definition 1.4. A coordinate chart is an injective map $\varphi : U \to \mathbb{R}^n$ with open image for some $U \subset M$.

In particular, we say that two charts $\varphi : U \to \mathbb{R}^n$ and $\psi : V \to \mathbb{R}^n$ are compatible if the transition $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is a diffeomorphism. Clearly, if $U \cap V = \varnothing$, then said maps are compatible; note, however, that compatibility is not an equivalence relation, since, while it is symmetric and reflexive, it is not necessarily transitive.

§2 Day 2: Atlases (Jan. 8, 2026)

We correct an error from last class. Let M be a set, and consider $\varphi : U \to \mathbb{R}^n$, where $U \subset M$. We call φ a (coordinate) chart if $\varphi(U)$ is open and φ is injective; indeed, we identify U with an open subset of $\mathbb{R}^n$. Given two charts $\varphi : U \to \mathbb{R}^n$ and $\psi : V \to \mathbb{R}^n$, we say they are *compatible* if

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$$

is a diffeomorphism of open subsets.[2] We now present an idea; indeed, we may regard φ as a coordinate system on U, i.e., $\varphi(p) = (x^1, \ldots, x^n)$ are *coordinates* of p (with respect to φ), for which our transition maps are realized as coordinate changes.

Definition 2.1. A set of charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in I}$ that covers M (i.e., every $p \in M$ is in some U_α) is called an *atlas*.

From this we may draft the following definition,

Definition. A *manifold* is a set M with an atlas $\mathcal{A}$ of charts.

Observe the following examples,

(i) Consider the n-sphere S^n, for which we have the stereographic projection φ_N, projecting N through a point on the sphere onto $\mathbb{R}^n$, which we may define as follows,

$$\varphi_N(x_1, \ldots, x_{n+1}) = \frac{1}{1 - x_{n+1}}(x_1, \ldots, x_n).$$

However, it is evident that φ_N does not admit N in its domain; thus, we may similarly define the stereographic projection from the south pole by

$$\varphi_S(x_1, \ldots, x_{n+1}) = \frac{1}{1 + x_{n+1}}(x_1, \ldots, x_n)$$

in order to cover S^n. Do these two maps form an atlas for S^n? It suffices to check that they are compatible; directly write as follows,[3]

$$\varphi_S \circ \varphi_N^{-1}(y_1, \ldots, y_n) = \frac{1}{|\vec{y}|^2}(y_1, \ldots, y_n),$$

where $\vec{y} = (y_1, \ldots, y_n)$.[4]

(ii) Let M be the set of straight lines in $\mathbb{R}^2$, i.e., of the form "$ax + by = c$". Consider the charts φ mapping $mx + b$ to $(m, b) \in \mathbb{R}^2$ and ψ mapping $x = ny + c$ to $(n, c) \in \mathbb{R}^2$ (in this manner, we account for both horizontal and vertical lines). Then the transition map between them is realized as

$$\psi \circ \varphi^{-1}(m, b) = \left(\frac{1}{m}, -\frac{b}{m}\right)$$

whenever $m \neq 0$ (when the line is neither horizontal nor vertical).

Given two atlases, how do we test that they define the same manifold?

[2] defn 2.4 in gross meinrenken

[3] see: p.22-23 for the S^1 case

[4] fedya u seem cool n all but we cannot be friends if u use this notation (that is, until i change my mind and start appreciating $\vec{\bullet}$)

Definition 2.2 (§2.8)**.** A chart $\varphi : U \to \mathbb{R}^n$ is *compatible* with an atlas if $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ if φ is compatible with every φ_α.

Lemma 2.3 (§2.10)**.** If $\varphi : U \to \mathbb{R}^n$, $\psi : V \to \mathbb{R}^n$ are compatible with the same atlas $\mathcal{A}$, then they're compatible with each other.

Proof. It is straightforward to check the properties, so just read the book. □

Theorem 2.4 (§2.11)**.** Given an atlas $\mathcal{A}$ on M, there's a unique *maximal* atlas $\widetilde{\mathcal{A}}$ which consists of all charts compatible to $\mathcal{A}$. Every chart compatible with $\mathcal{A}$ is already in $\widetilde{\mathcal{A}}$.

Proof. In one direction, if a chart is compatible with $\tilde{\mathcal{A}}$, then it is compatible with $\mathcal{A}$, and is therefore in $\widetilde{\mathcal{A}}$; in the other direction, $\widetilde{\mathcal{A}}$ is an atlas, so it covers because it contains $\mathcal{A}$ and is pairwise compatible by the lemma, so we are done. □

With this, we give a second attempt and defining a manifold.

Definition. A manifold is a set M with a maximal atlas $\mathcal{A}$.

§3 Day 3: Real and Complex Projective Plane; Real and Complex Grassmannians (Jan. 13, 2026)

Last class, we finished with our "second draft" of a definition of a manifold, i.e., a set M with a maximal atlas $\mathcal{A}$ (if a chart is compatible with $\mathcal{A}$, then it is already in $\mathcal{A}$). In practice, it is sufficient to take a "reasonably large" atlas, from which it can be uniquely completed to a maximal atlas. Two atlases generate the same maximal atlas if and only if each chart from one atlas is compatible with every chart in the other.

There are still some problems with our definition, though. For example, if M is "too big", such as $M = \mathbb{R}$ and we consider the atlas $\varphi_r : U_r \to \mathbb{R}^0 = \{0\}$, where each $U_r = \{r\} \in \mathbb{R}$, then M is just "uncountable dust", and doesn't fit in every $\mathbb{R}^N$.[5]

Definition 3.1. A manifold $(M, \mathcal{A})$ is *second countable* if it can be covered by countably many charts.

As another example, consider the line with two origins, $\mathbb{R} \cup \{0'\}$; we may give this two charts, $\mathrm{id} : \mathbb{R} \to \mathbb{R}$, and $\mathbb{R} \setminus \{0\} \cup \{0'\} \to \mathbb{R}$ given by the identity everywhere except $0' \mapsto 0$.

Definition 3.2 (§2.16b)**.** A manifold $(M, \mathcal{A})$ is *Hausdorff* if every pair of points lie in disjoint charts, i.e., for all $p_1, p_2 \in M$, there are

$$\varphi_1 : U_1 \to \mathbb{R}^n, \quad \varphi_2 : U_2 \to \mathbb{R}^n,$$

such that $U_1 \cap U_2 = \varnothing$, $p_1 \in U_1$, and $p_2 \in U_2$. Note that in a maximal atlas, given a chart $\varphi : U \to \mathbb{R}^n$, the restriction

$$\varphi\big|_V : V \to \mathbb{R}^n$$

is also a chart as long as $\varphi(V)$ is open.

In particular, this means that if p_1, p_2 are in the same chart, they can't obstruct the Hausdorff property. With this, we can define the manifold properly,

Definition 3.3. A manifold is a set M with a maximal atlas $\mathcal{A}$ that is second countable and Hausdorff.

We now present some examples.

(a) (§2.18) Let S^n be equipped with the maximal atlas containing the stereographic projections φ_N and φ_S. M is second countable because it is covered by two charts, and Hausdorff because the only interesting case to look at is when $p_1 = N$ and $p_2 = S$ (otherwise, we can cover p_1, p_2 with the same chart), but we can then restrict φ_S and φ_N on the open hemispheres.

(b) (Section §2.3.2) Consider the real projective n-space $\mathbb{R}\mathrm{P}^n$, which can be regarded as the set of lines through 0 in $\mathbb{R}^{n+1}$, or

$$(\mathbb{R}^{n+1} \setminus \{0\}) / \big\{v \sim cv \mid c \in \mathbb{R} \setminus \{0\}\big\},$$

or $S^n / v \sim -v$ (read: equivalence classes), or $\overline{B^n} / \{v \sim -v \mid v \in \partial \overline{B_n}\}$, where $B^n = \{x \in \mathbb{R}^n \mid \|x\|^2 \leq 1\}$. Note that all of these ways to express $\mathbb{R}\mathrm{P}^n$ are equivalent to each other.

[5] this is example §2.14 in the textbook, p.26

Definition 3.4. Since any nonzero vector in $\mathbb{R}^{n-1}$ determines a line (through the origin), we write $(x^0 : x^1 : \cdots : x^n)$ to be the *homogeneous coordinates* in the equivalence class of $(x^0, \ldots, x_n) \in \mathbb{R}^{n-1} \setminus \{0\}$ with equivalence if two points agree up to a nonzero scalar multiple.

The projective space $\mathbb{R}\mathrm{P}^n$ can be given a chart, i.e., a projection taking the open northern hemisphere to $\mathbb{R}^n$ by

$$\varphi : (x^0 : x^1 : \cdots : x^n) \mapsto \left(\frac{x^1}{x^0}, \ldots, \frac{x^n}{x^0}\right),$$

for which φ_0 admits the domain $U_0 := \{(x^0 : x^1 : \cdots : x^n) \mid x^0 \neq 0\}$. We may, in this manner, construct $\varphi_1, \ldots, \varphi_n$ similarly covering $\mathbb{R}\mathrm{P}^n$ (where we omit the jth coordinate for φ_j)[6], since every point in $\mathbb{R}\mathrm{P}^n$ has some nonzero homogeneous coordinate. In this manner, we may view U_j to consists of the lines intersecting the affine hyperplane $H_j = \{x \in \mathbb{R}^{n+1} \mid x^j = 1\}$.

Exercise 3.5. Compute the transition maps for $\mathbb{R}\mathrm{P}^n$ and verify that they're smooth. What do we need in order to show that this is a manifold? We need to show that (i) the charts cover $\mathbb{R}\mathrm{P}^n$, (ii) the transition maps are smooth (which induces some maximal atlas), (iii) is second countable, (iv) and is Hausdorff. Note that the first two conditions are shown because we already demonstrated earlier that the $\varphi_0, \ldots, \varphi_n$ form a chart, and that second countability comes from (i). We will show (iv) later.

(c) (Section §2.3.3) We now discuss the complex projective space $\mathbb{C}\mathrm{P}^n$, which we take to be $\mathbb{C}^{n+1} \setminus \{0\}$ quotientied out by constant multiples, or S^{2n+1} quotiented out by the relation $v \sim e^{i\theta}v$, where $\theta \in [0, 2\pi)$ (each equivalence class is a circle). Here, we write the homogeneous coordinates to be $(z^0 : z^1 : \cdots : z^n)$, and we have a similar construction for charts,

$$\varphi_i(z^0 : z^1 : \cdots : z^n) = \left(\frac{z^0}{z^i}, \ldots, \frac{z^{i-1}}{z^i}, \frac{z^{i+1}}{z^i}, \ldots, \frac{z^n}{z^i}\right),$$

where $\varphi_i : U_i \to \mathbb{R}^{2n} \cong \mathbb{C}^n$. As an example, consider $\mathbb{R}\mathrm{P}^1 \cong \mathbb{R} \cup \{(1 : 0)\} \cong S^1$, and $\mathbb{C}\mathrm{P}^1 \cong \mathbb{C} \cup \{(1 : 0)\} \cong S^2$, where we view $\mathbb{C} = \{(z : 1) \mid z \in \mathbb{C}\}$. Note that this is not a rigorous visualization.

Remark 3.6. The transition maps here are not just diffeomorphic, but also holomorphic. $\mathbb{C}\mathrm{P}^n$ can be regarded as a complex manifold, and in general, you can define manifolds with some "extra structure" if their transition maps have said extra structure.

(d) (Section §2.3.4) The *real Grassmannian* is denoted $\mathrm{Gr}(k, n)$ and defined as the k-dimensional vector subspaces of $\mathbb{R}^n$. As an example,

$$\mathbb{R}\mathrm{P}^n = \mathrm{Gr}(1, n+1) \cong \mathrm{Gr}(n, n+1).$$

How do we build charts in this manifold? As an idea, we can think of the subspaces as graphs of linear functions; one such chart might be

$$\operatorname{span}\begin{pmatrix} 1 & 0 & a \\ 0 & a & b \end{pmatrix} \to (a, b) \in \mathbb{R}^2;$$

in general, there are $\binom{n}{k}$ total charts[7].

[6] see p.31 for a more careful treatment

[7] i will leave some of the exposition to the textbook, i don't like what fedya is doing

Claim 3.7. These charts cover $\mathrm{Gr}(k, n)$, i.e., in each k-dimensional subspace E, we can find $e_{i_1}, \ldots, e_{i_k}$, such that E is the graph of a function sending $(x^{i_1}, \ldots, x^{i_k})$ to the other $n - k$ coordinates in $\mathbb{R}^n$.

Since I've basically lost what Fedya is saying because he writes in cursive and his jumping around everywhere in the textbook treatment for real Grassmannians, I direct you to read pages 34-36 for the same treatment of this subject that has already been written down.

§4 Day 4: Topology on Manifolds (Jan. 20, 2026)

Given a manifold M with a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$, we define a subset $U \subset M$ to be open if, for all charts $\varphi_\alpha \colon U_\alpha \to \mathbb{R}^n$, $\varphi_\alpha(U \cap U_\alpha)$ is open in $\mathbb{R}^n$.

Proposition 4.1. U is open if and only if $\varphi_\beta(U \cap U_\beta)$ is open in (U_β, φ_β) covering U (e.g., a non-maximal atlas).

For example, $U \subset S^n$ is open if and only if $\varphi_N(U \setminus \{N\})$ and $\varphi_S(U \setminus \{S\})$ is open.

Proof. The forward direction is clear, so we will check the converse only. Let $\varphi_\alpha \colon U_\alpha \to \mathbb{R}^n$ be another chart in $\mathcal{A}$; then $U \cap U_\alpha = \bigcup_\beta U \cap U_\alpha \cap U_\beta$, whence

$$\begin{aligned}\varphi_\alpha(U \cap U_\alpha \cap U_\beta) &= (\varphi_\alpha \circ \varphi_\beta^{-1}) \circ \varphi_\beta(U \cap U_\alpha \cap U_\beta) \\ &= (\varphi_\alpha \circ \varphi_\beta^{-1})(\varphi_\beta(U \cap U_\beta) \cap \varphi_\beta(U_\alpha \cap U_\beta)).\end{aligned}$$

Indeed, each $\varphi_\beta(U \cap U_\beta)$ is open by assumption and each $\varphi_\beta(U_\alpha \cap U_\beta)$ is open because φ_α and φ_β are compatible charts (as they're in the same atlas). Thus, $\varphi_\beta(U \cap U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n$, and so $\varphi_\alpha(U \cap U_\alpha \cap U_\beta)$ is open, so $\varphi_\alpha(U \cap U_\alpha)$ is a union of open sets and is therefore itself open. □

Fact 4.2. If $\mathcal{A}$ is an atlas on M and $U \subset M$ is open, then $A_U = \{(U \cap U_\alpha, \varphi_\alpha|_{U \cap U_\alpha})\}$ is an atlas on U. The proof of this fact is left as an exericse.

Moreover, observe that if $\mathcal{A}_U$ inherits maximality, second countability, and Hausdorff-ness from $\mathcal{A}$, meaning that if $(M, \mathcal{A})$ is a manifold, then $(U, \mathcal{A}_U)$ is one as well.

Proposition 4.3. Using our definition from earlier, the collection of open subsets of M forms a topology on M. Specifically, $\varnothing, M$ are open, the finite intersection of open sets are open, and the arbitrary union of open sets is also open.

The proof sketch is that we will use the fact that the above is indeed true in $\mathbb{R}^n$, so we will adapt that idea down here. We start by defining more things in topology, though. We call $N \subset M$ a neighborhood of p if there exists open U such that $p \in U \subset N$. A set A is *closed* if its complement is open. M is *disocnnected* if it can be written as $U \sqcup V$, where U, V are open and nonempty (equivalently, this means there are no nonempty proper subsets of M that are clopen).

For example, $S^0 = \{-1, 1\} \subset \mathbb{R}$ has $\{-1\}$ and $\{1\}$ as open sets, so they are both disconnected. We say M is connected if it is not disconnected. We say M is Hausdorff if any two points have disjoint neighborhoods.

Remark 4.4. U is open if and only if it's a union of U_α, where $(U_\alpha, \varphi_\alpha)$ are charts in a maximal atlas. (A maximal atlas gives a basis for the manifold topology).

Proof. If $(U_\alpha, \varphi_\alpha)$ is a chart and $\varphi(U \cap U_\alpha)$ is open, then $(U \cap U_\alpha, \varphi_\alpha|_{U \cap U_\alpha})$ is also a chart in the maximal atlas. □

We now discuss compactness. $K \subset \mathbb{R}^n$ is compact if it is closed and bounded (for example, a closed ball).

Definition 4.5. $K \subset M$ is compact if $K = K_1 \cup \cdots \cup K_n$, where $K_i \subset U_i$ and $\varphi_i(K_i) \subset \mathbb{R}^n$ is compact.

In particular, in $\mathbb{R}^n$, finite unions of compact sets are compact. Alternatively, we can define K to be compact if $K \subset U_i$ and $\varphi_i(K)$ is compact in $\mathbb{R}^n$. We give some examples.

(i) S^n is compact. Moreover, if we take $\{x \in S^n \mid x_{n+1} \geq 0\}$ and $\{x \in S^n \mid x_{n+1} \leq 0\}$ to be the upper and lower hemispheres of S^n, then stereographic projection from the opposite poles will demonstrate that their images are compact in $\mathbb{R}^n$.

(ii) $\mathbb{R}\mathrm{P}^n$ is compact, since we may regard it as $\{[x_0 : x_1 : \cdots : x_n]\}$ with an atlas $U_i = \{[x_0 : x_1 : \cdots : x_n] \mid x_i \neq 0\}$ and its associated map φ_i; by noticing that if we manually construct $x_0^2 + \cdots + x_n^2 = 1$, there is some element $x_i \geq \sqrt{n}^{-1}$, so $\|\varphi_i([x_0 : x_1 : \cdots : x_n])\| \leq \sqrt{n}$, i.e.,

$$\mathbb{R}\mathrm{P}^n = \bigcup_i \varphi_i^{-1}\big(\overline{B_{\sqrt{n}}(0)}\big).$$

In topology, we have the following definition for compactness.

Definition 4.6. K is compact if, for all collections $\{U_\alpha\}$ of open sets such that $\bigcup_\alpha U_\alpha \supset K$ (i.e., the U_α cover K), there is a finite subcollection that also covers K.

Proposition 4.7. If K is topologically compact in a manifold M, then K is closed. If $C \subset M$ is closed, then $K \cap C$ is compact.

Proof. Suppose K is not compact; then $M \setminus K$ is not open, so there exists a point $y \notin K$ such that every open neighborhood of y intersects K. By Hausdorffness, for all x, there exists $U_x \ni x$ and $V_x \ni y$ an open disjoint set such that $\{U_x \mid x \in K\}$ covers K. This cover has a finite subcover, for which $K \subset U_{x_1} \cup \cdots \cup U_{x_n}$ and $K \cap (V_{x_1} \cap \cdots \cap V_{x_n}) = \varnothing$, which contradicts our assumption on y.

For the second part of the proposition, take $\{U_\alpha\}$ an open cover of $K \cap C$; then $\{U_\alpha\} \cup \{M \setminus C\}$ is an open cover of K. By compactness, this has a finite subcover for K; taking out $M \setminus C$, we get a finite subcover of $\{U_\alpha\}$ for $K \cap C$. $\square$

Proposition 4.8. The two definitions for compactness are equivalent. Note that we assume this fact for $\mathbb{R}^n$ (per the Heine–Borel theorem).

Proof. Suppose K is charts-compact; then $K = K_1 \cup \cdots \cup K_n$, where $k_i \subset U_i$ for some charts (U_i, φ_i). Let $\{V_n\}$ be an open cover of K; by Heine–Borel, each V_i is covered by finitely many $\varphi_i(V_\alpha \cap U_i)$, so K admits a finite subcover of $\{V_n\}$.

For the other direction, suppose K is topologocaily compact. Take the open cover $\{V_{U,\varphi,x,\varepsilon}\}$, where $V_{U,\varphi,x,\varepsilon} = \varphi^{-1}(B_\varepsilon(x))$ where U, φ is a chart, $x \in \varphi(U)$, and $\overline{B_\varepsilon(x)} \subset \varphi(U)$. Take a finite subcover $V_{U_i,\varphi_i,x_i,\varepsilon_i}$. Then

$$K \subset \bigcup_{i=1}^n \varphi_i^{-1}(B_{\varepsilon_i}(x_i)) \implies K = \bigcup_{i=1}^n \varphi_i^{-1}(\overline{B_{\varepsilon_i}(x_i)} \cap K).$$

By the earlier proposition, this set is compact, so we are done. $\square$

§5 Day 5: Orientability (Jan. 22, 2026)

Observe that the Möbius strip (following our construction in this class) only has "one side". We can associate the right hand rule with $\det(v_1 \mid v_2 \mid v_3) > 0$.

Definition 5.1. A linear transformation $T\colon V \to V$ is *orientation preserving* if $\det T > 0$; a transition map $\tau\colon V_1 \to V_2$ between two charts is *orentation preserving* if $\det(D\tau) > 0$ everywhere.

Exercise 5.2. If $\det(D\tau) > 0$ in one half of the Möbius strip, then it has $\det(D\tau) < 0$ in the other half, and vice versa.

Definition 5.3. We say φ_1 and φ_2 are *orientation compatible* if their transition map is orientation preserving.

Definition 5.4. An *oriented atlas* is one in which all charts are orientation compatible. A *maximal oriented atlas* contains all charts orientation compatible with a given oriented manifold $(M, \mathcal{A})$.

Note that the definition of maximal oriented atlas only makes senes if, in any oriented atlas, any two charts that are oriented compatibly to said atlas are orientation compatible to each other. This is left as an exercise. We now give examples.

(i) The sphere is orientable. We claim that the usual atlas $\{\varphi_N, \varphi_S\}$ is not oriented compatibly, but $\{\varphi_N, (-x_1, x_2, \ldots, x_n) \circ \varphi_S\}$ is. Observe that for $n \geq 2$, the intersection $U_N \cap U_S = S^n \setminus \{N, S\}$ is connected, meaning that the sign of $\det(D\tau) \neq 0$ on the whole set (by the intermediate value theorem).

Fact 5.5. Let X be a connected space, and let $f\colon X \to A$ (where A is taken to be some set) be locally constant, i.e., for every $p \in X$, there exists an open neighborhood $U \ni x$ such that $f|_U$ is constant. Then f is constant.

Proof. Observe that the preimage of fibers $f^{-1}(a) = \bigcup_{p \in f^{-1}(a)} U_p$ is open, so X can be realized as the disjoint union $\bigsqcup_{a \in A} f^{-1}(a)$; per connectedness of X, it must be that there is only one such a. □

(ii) $\mathbb{R}\mathrm{P}^2$ is not orientable. We leave this as an exercise.

Definition 5.6. Let $(M, \mathcal{A})$ be an oriented manifold. Then there is an opposite orientation $\tilde{\mathcal{A}} = \{F(U), F \circ \varphi\}$ for each $(U, \varphi) \in \mathcal{A}$, where $F\colon \mathbb{R}^n \to \mathbb{R}^n$ and $x \mapsto (-x_1, x_2, \ldots, x_n)$.

Proposition 5.7. Let $(M, \mathcal{A})$ be an oriented manifold. Every connected chart (U, φ) is oriented compatible with $\mathcal{A}$ or $\tilde{\mathcal{A}}$.

Proof. Let $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$, and define $\Sigma_p \coloneqq \operatorname{sign}(\det D(\varphi_\alpha \circ \varphi^{-1})\big|_{\varphi(p)})$ is independent of α. We may note that $\Sigma_p\colon U \to \{+, -\}$ is a locally constant function; but U is connected, so it must be constant. □

We leave it as an exercise that if M is a connected manifold, then there is exactly one maximal oriented atlas.

§6 Day 6: Smooth maps (Jan. 27, 2026)

We know that charts and their inverses are smooth by definition, and compositions of smooth functions are smooth. We wish to define a general notion of smooth maps between manifolds. Let us start with a first notion; a function $f\colon M \to \mathbb{R}$ is *smooth* at $p \in M$ if, for some chart (U, φ), $p \in U$, we have that

$$f \circ \varphi^{-1}\colon \varphi(U) \to \mathbb{R}$$

is smooth, i.e., C^∞ at $\varphi(p)$. φ is smooth, $f \circ \varphi^{-1}$ is smooth, so $f = (f \circ \varphi^{-1}) \circ \varphi$ is smooth.

Remark 6.1. Smoothness is independent of the choice of charts, since transition maps are smooth, so we can replace "some chart" with "every chart" in the definition above.

A function $f\colon M \to \mathbb{R}$ is *smooth* if it is smooth at every point; equivalently, for every chart (U, φ), $f \circ \varphi^{-1}$ is smooth. Also equivalently, for a set of charts (U, φ) covering M, we have that $f \circ \varphi^{-1}$ is smooth. This is the easiest condition to check. We give some examples.

(i) $f\colon \mathbb{R}^{n+1} \to \mathbb{R}$ is smooth. Then its restriction to $S^n \subset \mathbb{R}^{n+1}$ is also smooth. We will prove this via stereographic projection φ_N, φ_S. Indeed, $\varphi_N^{-1}, \varphi_S^{-1}\colon \mathbb{R}^n \to \mathbb{R}^{n+1}$ is smooth, so $f \circ \varphi_N^{-1}$ and $f \circ \varphi_S^{-1}$ are both smooth as well, meaning the restriction of f down to S^n is smooth as desired.

(ii) Let $f\colon S^2 \to \mathbb{R}$ with $f(x, y, z) = \sqrt{1 - z^2} = \sqrt{x^2 + y^2}$. We can use the chart $\varphi\colon \{(x, y, z) \mid z > 0\} \to B_1(0)$ by $\varphi(x, y, z) = (x, y)$, where, $f \circ \varphi^{-1}(x, y) = \sqrt{x^2 + y^2}$ is smooth at the origin, so f is not smooth.

We denote $C^\infty(M) = \{f\colon M \to \mathbb{R} \mid f \text{ is smooth}\}$ to be the collection of smooth functions on M.

Fact 6.2. Linear combinations of $f, g \in C^\infty(M)$ is smooth; the pointwise product is also smooth, and $1 \in C^\infty(M)$ is the multiplicative identity. Thus, $C^\infty(M)$ forms an $\mathbb{R}$-algebra.

We now discuss continuity. $f\colon M \to \mathbb{R}$ is continuous if and only if $f \circ \varphi^{-1}$ is continuous for all charts (U, φ). In particular, smooth functions are continuous.

Proposition 6.3. This definition of continuity is equivalent to the topological notion of continuity.

Proof. By the same fact for $U \subset \mathbb{R}^n$, we have that $(f \circ \varphi_\alpha^{-1})^{-1}(a, b)$ is open in $\varphi_\alpha(U_\alpha) \subset \mathbb{R}^n$ for every chart $(U_\alpha, \varphi_\alpha)$ Recall that $V \subset M$ is open if and only if it is a union of $\varphi_\alpha^{-1}(V_\alpha)$, where V_α is open in $\varphi_\alpha(U_\alpha)$. This means

$$f^{-1}(a, b) = \bigcup_\alpha \varphi_\alpha^{-1}\big((f \circ \varphi_\alpha^{-1})^{-1}(a, b)\big)$$

is open. $\square$

We now discuss supports. The support of $f\colon M \to \mathbb{R}$ is the *closure* of $f^{-1}(\mathbb{R} \setminus \{0\})$. If $p \notin \operatorname{supp}(f)$, then f is 0 in a neighborhood of p.

Lemma 6.4. If $U \subset M$ is open, $g\colon U \to \mathbb{R}$ is smooth, and $\operatorname{supp}(g) \subset U$ is closed in M, then g extends to a smooth $f\colon M \to \mathbb{R}$, $f|_U = g$, and $f|_{M \setminus U} = 0$.

Proof. Let $V = M \setminus \operatorname{supp}(g)$. This is open in M. f is smooth at $p \in U$ (take any chart containing p with domain in U) and $p \in V$ (take any chart containing p with domain in V and coordinate chart equal to the zero map); then $M = U \cup V$, so f is smooth everywhere on M. □

Proposition 6.5. A set $(M, \mathcal{A})$ with a *maximal atlas* is *Hausdorff* if and only if, fo rall $p \neq q \in M$, there exists a smooth $f\colon M \to \mathbb{R}$ such that $f(p) \neq f(q)$.

Proof. In the reverse direction, given $p \neq q$, we want to show that they lie in disjoint open sets. Choose $f\colon M \to \mathbb{R}$ such that $f(p) \neq f(q)$; then $f^{-1}(f(p) - \varepsilon, f(p) + \varepsilon)$ are disjoint open sets containing p and q (we may take $\varepsilon < \frac{1}{2}\,|f(p) - f(q)|$).

In the forwards direction, for any $p \neq q$, let U, V be disjoint open neighborhoods respectively, and let (W, φ) be a chart with $W \ni p$. Choose a smooth function $f\colon f(W) \to \mathbb{R}$ such that $f(\varphi(p)) = 1$ and f is supported on $B_\varepsilon(p)$. This is called a *bump function*. We have that $f \circ \varphi\colon W \to \mathbb{R}$ is smooth, so by the lemma, we can extend it to $g\colon M \to \mathbb{R}$ with $g(q) = 0$ and $g(p) = 1$. □

Corollary 6.6. Suppose $(M, \mathcal{A})$ is a set with a *maximal atlas*. Then, if there is a coordinate-wise smooth injection $f\colon M \to \mathbb{R}^N$, then M is Hausdorff.

Proof. For any $p \neq q$, $f(p) \neq f(q)$, so one of their coordinates is different, which gives us our separating function. □

Note that we do need Hausdorffness in order to be embedded in $\mathbb{R}^n$. As an example, consider $F\colon \mathbb{R}\mathrm{P}^n \to M_{(n+1)\times(n+1)}(\mathbb{R}) \cong \mathbb{R}^{(n+1)^2}$, given by

$$\left(x^0 : x^1 : \cdots : x^n\right) \mapsto \frac{xx^\top}{\|x\|^2}.$$

This map is well defined, smooth (coordinate wise), and injective (as the diagonal of the matrix is x_i^2), so $\mathbb{R}\mathrm{P}^n$ is Hausdorff.

We now discuss smooth maps between manifolds. $f\colon M \to N$ is smooth, if, for every chart of M and N, given (U, φ) on M and (V, ψ) on N, $\psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \to \psi(V)$ is smooth. We denote $C^\infty(M, N)$ as the collection of such smooth maps. We give an example.

(i) There are diffeomorphisms from $\mathbb{R}\mathrm{P}^1 \to S^1$ and $\mathbb{C}\mathrm{P}^1 \to S^1$. Indeed, take $(x : 1) \mapsto x \in \mathbb{R}$ then precompose with φ_N^{-1} to get a map into $S^1 \setminus \{N\}$. For the last point not of the form $(x : 1)$, we just take $(0 : 1) \mapsto N$. The same method holds for the complex projective space.

§7 Day 7: Smooth Maps and Diffeomorphisms (Jan. 29, 2026)

Recall that a map between two manifolds $f\colon M \to N$ is *smooth* at a point $p \in M$ if there are coordinate charts (U, φ) about p and (V, ψ) about $f(p)$ such that $f(U) \subset V$ and

$$\psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) = \varphi(U) \to \psi(V)$$

is smooth. f is said to be smooth if it is smooth at every $p \in M$, and it is said to be a diffeomorphism if it's a smooth bijection with smooth inverse. Last time, we gave an example of a diffeomorphism from $\mathbb{R}\mathrm{P}^1 \to S^1$ by considering $[x:1] \mapsto \varphi_N^{-1}(x)$ and $[1:0] \mapsto N$. We now give more examples.

(i) (§3.6.3) Let $\pi\colon \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}\mathrm{P}^n$, given by $(x^0, \ldots, x^n) \mapsto \left[x^0 : \cdots : x^n\right]$. It is clear that this map is smooth because

$$\pi \circ \varphi_i(x^0, \ldots, x^n) = \pi(\left[x^0 : \cdots : x^n\right]) = \left(\frac{x^0}{x^i}, \ldots, \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, \ldots, \frac{x^n}{x^i}\right).$$

(ii) Let $\pi\colon \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{C}\mathrm{P}^n$, where $(z^0, \ldots, z^n) \mapsto \left[z^0 : \cdots : z^n\right]$. The map is smooth by the same proof as above.

(iii) (§3.22) The map $\mathbb{C}\mathrm{P}^1 \to S^2$ given by $[z:1] \mapsto \varphi_N^{-1}(x, y)$ and $[1:0] \mapsto N$ is smooth. To see this, let $p = [z:1] \in \mathbb{C}\mathrm{P}^1$; we have the following diagram,

$$\begin{tikzcd}
\mathbb{C}\mathrm{P}^1 \setminus \{[1:0]\} \arrow[r, "f"] \arrow[d, "\varphi_1"] & S^2 \arrow[d, "\varphi_N"] \\
\varphi_1(\mathbb{C}\mathrm{P}^1 \setminus \{[1:0]\}) \arrow[r, "\mathrm{id}"] & \varphi_N(S^2 \setminus \{N\})
\end{tikzcd}$$

which clearly commutes, so we only have $p = [1:0]$ left to check. Indeed, swap out φ_1 with φ_0 and φ_N with φ_S; we get

$$\varphi_0([1:0]) = 0 = \varphi_S(N).$$

It suffices to check that we have smoothness in a neighborhood. In one direction, we have

$$\varphi_0([z:1]) = \varphi_0(\left[1 : z^{-1}\right]) = \frac{1}{2} = \frac{\overline{z}}{z^2};$$

in the other, we have

$$\varphi_S \circ \varphi_N^{-1}(x, y) = \frac{(x, y)}{x^2 + y^2} = \frac{z}{|z|^2},$$

so we see the map $\varphi_0(\mathbb{C}\mathrm{P}^1 \setminus \{[0:1]\}) \to \varphi_S(S^2 \setminus \{S\})$ is given by complex conjugation.

(iv) $f\colon \mathbb{R} \to \mathbb{R}$, where $f(x) = x^3$ is smooth and bijective, but $f^{-1}(x) = x^{1/3}$ is not differentiable at $x = 0$, so f is not a diffeomorphism!

We now check that compositions of smooth functions are smooth. A *homeomorphism* is a continuous bijection with continuous inverse (topological equivalence). For example, $x \mapsto x^3$ is a homeomorphism but not a diffeomorphism; are homeomorphic manifolds always diffeomorphic? In 1 to 3 dimensions, this is true, but otherwise it is not true.[8]

[8] yes bro drop us the paper bro

There are infinitely many spaces taht are homeomorphic but not diffeomorphic to $\mathbb{R}^1$.[9]

We now discuss Hopf fibration (§3.7). Specifically, consider the map $h\colon S^3 \to \mathbb{C}\mathrm{P}^1$ realized as the sequence

$$S^3 \hookrightarrow \mathbb{R}^4 \setminus \{0\} \xrightarrow{\pi} \mathbb{C}\mathrm{P}^1 \cong S^2.$$

What can we say? First this is smooth by composition, and given $p = [z : w] \in S^2$ (where, without loss of generality, we can assume $|z|^2 + |w|^2 = 1$), we see that $h^{-1}(p)$ is actually a circle

$$h^{-1}([z : w]) = \{(e^{i\theta} z, e^{i\theta} w)\} \subset S^3.$$

Recall that S^3 can be realized as the one point compactification of $\mathbb{R}^3$ (by considering the stereographic projection), i.e.,

$$h^{-1}\,(0 : 1) = (0, 0, t), \quad h^{-1}\,(1 : 0) = (\cos\theta, \sin\theta, 0).$$

In general,

$$h^{-1}\,(z : r) = \left(\frac{e^{i\theta} z}{1 - (\sin\theta) r}, \frac{(\cos\theta) r}{1 - (\sin\theta) r} \right)$$

[9] freedman, donaldson, ...

§8 Day 8: (Feb. 3, 2026)

§9 Day 9: Normal form theorem (Feb. 5, 2026)

Last class, we defined submanifolds, the rank of a smooth map, and maps of maximum rank (submersions) $f\colon M^m \to N^n$, where M is of dimension m and N is of dimension n with $m \geq n$.

Fact 9.1. In a submersion f, $f^{-1}(p)$ are submanifolds.

Recall that, if $f\colon M^m \to N^n$ with $m \leq n$ has rank m everywhere, then it is an *immersion*.

Theorem 9.2 (Normal form theorem)**.** Let $f\colon M \to N$ be a map of rank m at $p \in M$. Then there are charts (U, φ) at $p \in M$ and (V, ψ) at $f(p)$ for N such that the diagram commutes,

$$\begin{tikzcd} U \arrow[r, "f"] \arrow[d, "\varphi"] & V \arrow[d, "\psi"] \\ \varphi(U) \arrow[r, hook] & \psi(V) \end{tikzcd}$$

Proof. This follows from the $\mathbb{R}^n$ version. □

Equivalently, $f|_U$ is the inclusion of a submanifold, which we see by restricting (V, ψ) to $\psi^{-1}(U \times \mathbb{R}^{n-m})$. We now discuss examples.

(i) A curve is a smooth map $\gamma\colon (a, b) \to M$. A curve is immmersed at p if, for (V, ψ) a chart, $\frac{d}{dt}(\psi \circ \gamma)|_t \neq 0$. However, if we consider $\gamma(t) = (t^2, t^3)$ or (t^3, t^3) in $\mathbb{R}^2$, the image *is* a submanifold, but the parameterization is bad!

(ii) Consider the figure eight, given by the map $S^1 \to \mathbb{R}^2$.

(iii) Consider $\mathbb{R} \to \mathbb{R}^2$ given by $t \mapsto e^{it}$.

(iv) (§4.42) something something immersion to get a klein bottle.

Definition 9.3. An *embedding* is an injective immersion, where the image is a submanifold.

Theorem 9.4. If M is compact, then any injective immersion $f\colon M \to N$ is an embedding.

Proof. We want to show that, at each point $p \in M$, there is a submanifold chart (V, ψ), i.e.,

$$\begin{tikzcd} V \cap f(M) \arrow[r, hook] \arrow[d, "\psi|_M"] & V \arrow[d, "\psi"] \\ \psi(V) \cap \mathbb{R}^m \arrow[r] & \psi(V) \end{tikzcd}$$

To start, we know that there exists charts (U, φ) and (V, ψ) such that

$$\begin{tikzcd} U \arrow[r, "f"] \arrow[d, "\varphi"] & V \arrow[d, "\psi"] \\ \psi(V) \cap \mathbb{R}^m \arrow[r] & \psi(V) \end{tikzcd}$$

Recall that the images of compact sets are compact. This means $f(M)$ is compact in N, whence $f(M \setminus U) = A$ is closed (because closed subsets of compact sets are compact). Thus, $V \setminus A$ is open, and so $f(U) = (V \setminus A) \cap f(M)$. From this, we get the diagram

$$\begin{tikzcd} (V \setminus A) \cap f(M) \arrow[r] \arrow[d, "\psi|_{f(M)}"] & V \setminus A \arrow[d, "\psi"] \\ \psi(V) \cap \mathbb{R}^m \arrow[r] & \psi(V) \end{tikzcd}$$

$\square$

We give an example. Consider $A \neq B \neq C \in \mathbb{R}$; then define

$$F_{A,B,C} \colon \mathbb{R}\mathrm{P}^2 \to \mathbb{R}^4, \quad (\alpha : \beta : \gamma) \mapsto (\beta\gamma, \alpha\gamma, \alpha\beta, A\alpha^2, B\beta^2, C\gamma^2),$$

where $(\alpha, \beta, \gamma) \in S^2$, is an injective immersion, i.e., an embedding, since $\mathbb{R}\mathrm{P}^2$ is compact.

§10 Day 10: (Feb. 10, 2026)

§11 Day 11: Derivations on Tangent Spaces (Feb. 12, 2026)

For any curve $\gamma\colon (a,b) \to M$ on a manifold, we have the velocity vector

$$\frac{d\gamma}{dt}.$$

The derivatives of a smooth map $f\colon M \to N$ are given by[10] $Df_p\colon T_pM \to T_{f(p)}N$, which is a generalization of the notion of Jacobians of maps $\mathbb{R}^m \to \mathbb{R}^n$. Recall that, for all $v \in T_pM$, where $v\colon C^\infty(M) \to \mathbb{R}$, we say v is a *tangent vector* if it satisfies the product rule[11]

$$v(g) = \frac{d}{dt}\bigg|_{t=0} g(\gamma(t)),$$

where $\gamma\colon (a,b) \to M$ is a curve with $\gamma(0) = p$.

Definition 11.1 (§5.11)**.** $Df_p(v)(g) = v(g \circ f)$ for $g \in C^\infty(N)$.

Proposition 11.2. The above definition is equivalent to $Df_p([\gamma]) = [f \circ \gamma]$.

Proof. Directly write as follows,

$$Df_p(v)(h) = v(h \circ f) = \frac{d}{dt}\bigg|_{t=0} (h \circ f)(\gamma(t)) = \frac{d}{dt}\bigg|_{t=0} h(f \circ \gamma(t)) = [f \circ \gamma] \in T_{f(p)}N. \quad \square$$

We introduce some notation. We denote $Df_p(v)$ also as f_*v, i.e., the *pushforward* of v. We claim that the chain rule holds.

$$D(g \circ f)_p(v) = Dg_{f(p)} \circ Df_p(v).$$

Proof. Consider the diagram[12]

$$\begin{tikzcd}
T_pM \arrow[r, "Df_p"] \arrow[d, "D\varphi_p"] & T_{f(p)}N \arrow[d, "D\psi_{f(p)}"] \\
\mathbb{R}^m \arrow[r, "D(\psi\circ f\circ\varphi^{-1})_{\varphi(p)}"'] & \mathbb{R}^n
\end{tikzcd}$$

We may notice that $D(\psi \circ f \circ \varphi^{-1})_{\varphi(p)}$ is just the Jacobian on $\mathbb{R}^m \to \mathbb{R}^n$. $\square$

We now revisit a definition from last week. The rank of f at p is the rank of Df_p. f is a *submersion* if Df_p is surjective, immersion if it injective, and a critical point if $Df_p \leq \min\{m,n\}$. It is a regular point if $Df_p = \min\{m,n\}$. We now discuss tangent spaces of submanifolds. Given $S \subset M$ a submanifold, we have $\iota\colon S \to M$ as an injection, which induces the injection $T_pS \hookrightarrow T_pM$. As an exapmle, recall that if $f\colon M \to N$, if $q \in N$ is a regular value then $f^{-1}(q)$ is an $m-n$ submanifold in M.

Claim 11.3 (§5.18)**.** Let q be a regular value of $f\colon M \to N$; then if we denote $f^{-1}(q)$ to be the submanifold S in M, we have that $T_pS = \ker Df_p$.[13]

[10] check p.55 of lee's smooth manifolds or section 1.2 of guillemin and pollack's differential topology
[11] read: look up derivations for a treatment on this subject.
[12] there is a much better diagram for this holy shit fedya.
[13] check the proof inside meinrenken's book, or the proposition at p.24 at the bottom of guillemin and pollack, or insert thing from lee's smooth manifolds.

Proof. Both sides have dimension $m - n$, so it is enough to show that $T_pS \subset \ker Df_p$. The second observation is that $f \circ \iota \equiv q$, whence $Df_p \circ D\iota_p = D(f \circ \iota)_p = 0$, whence $T_pS \subset Df_p$. $\square$

Note that, if we have $T_pS \subset \mathbb{R}^{n+1}$ as an n-dimensional hypersurface in $\mathbb{R}^{n+1}$, we can regard $T_pS = \langle p \rangle^\perp$ geometrically. For the sphere S^n, we can regard it as the preimage of the norm map $F^{-1}(1)$, where $F(x) = \|x\|$. In particular, $\nabla_p F = 2p$, hence $T_pS^n = \operatorname{span}(p)^\perp$.

For a last application of tangent spaces, let G be a matrix Lie group (under matrix multiplication), regarded as a subgroup of $M_{n\times n}(\mathbb{R}) \cong \mathbb{R}^{n^2}$ (specifically, G admits the identity matrix I_n and is closed under matrix multiplication).

Definition 11.4. $T_IG \subset M_{n\times n}(\mathbb{R})$ is the Lie algebra of G, denoted $\mathfrak{g}$.

Fact 11.5. $\mathfrak{g}$ is closed under $[X, Y] = XY - YX$, i.e., the matrix commutator. This operation is called the Lie bracket.

Proof. Left as an exercise in the book. check g&p p.22 $\square$

§12 Day 12: Vector Fields, Tangent Bundles, and Vector Bundles (Feb. 24, 2026)

We start on chapters 6 and 9 now.[14] A vector field is a "smooth choice of tangent vector at every point in M"; for $M \subset \mathbb{R}^n$, the tangent space T_pM is in $\mathbb{R}^n$.

Definition 12.1. A vector field is a smooth map $V\colon M \to \mathbb{R}^n$ such that $V(p) \in T_pM \subset \mathbb{R}^n$.

(i) Consider $S^2 \subset \mathbb{R}^3$ (to be thought of here as a globe). We can consider the latitude and longitude lines to give us a vector field on S^2. Consider the map

$$(\varphi, \theta) \mapsto (\cos\theta\cos\varphi, \cos\theta\sin\varphi, \sin\theta),$$

and $\partial_{\text{long}}(\varphi,\theta) = \cos\theta(\sin\varphi, \cos\varphi, 0)$, $\partial_{\text{lat}}(\varphi,\theta) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. These are smooth maps. We also have that $\partial_{\text{lat}}(\varphi,\theta)$ is orthogonal to the vector p, so it is in T_pS^2. There's no way to extend this smoothly to the north pole, so it is not a vector field on S^2.[15]

What does "vector field" mean on an abstract manifold? One way to think about them is to loko at charts.

Definition 12.2. A vector field X on M is a function $p \in M$ to $X_p \in T_pM$ such that, for each chart $\varphi\colon U \to \mathbb{R}^n$, we have $D\varphi_p \circ X(p)\colon U \to \mathbb{R}^n$ is smooth (using the identification $T_pM \cong \mathbb{R}^n$, induced by the chart).

We may check that, if $\psi \circ V \to \mathbb{R}^n$ is another chart, then on $U \cap V$, $D\varphi \circ X$ is smooth if and only if $D\psi \circ X$ is smooth. In fact, $D(\varphi \circ \psi^{-1})$ transforms between the coordinates on T_pM induced by φ and ψ.

Definition 12.3. The *tangent bundle* TM of M^m is a smooth $2m$-manifold with a point set $\bigsqcup_{p\in M} T_pM$, (non-maximal) atlas $(\bigsqcup_{p\in U} T_pM, (\varphi\circ\pi, D\varphi_p)\colon \bigsqcup_{p\in U} T_pM \to \mathbb{R}^m \times \mathbb{R}^m)$ for charts (U, φ) in M. Note that the image of the charts on the tangent bundle are just $\varphi(U) \times \mathbb{R}^m$.

Remark 12.4. The tangent bundle comes with a projection $\pi\colon TM \to M$ by $T_pM \mapsto p$. Note that $\bigsqcup_{p\in U} T_pM$ from earlier is precisely just $\pi^{-1}(U)$. (Left as an exercise to check that π is smooth.)

Also, for each $p \in M$, the preimage $\pi^{-1}(p)$ (i.e., fiber) is a vector space. We also have local triviality, i.e., for each $p \in M$, there is a neighborhood $U \ni p$ such that the diagram commutes,

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "(D\varphi,\pi)"] \arrow[dr, "\pi"'] & & \mathbb{R}^m \times U \arrow[dl, "\pi"] \\
& U &
\end{tikzcd}$$

where $(D\varphi, \pi)$ is a smooth map and is linear on each $\pi^{-1}(\bullet)$. As an example, the two charts on S^2, $\varphi_N\colon S^2 \setminus \{N\} \to \mathbb{R}^2$ and $\varphi_S\colon S^2 \setminus \{S\} \to \mathbb{R}^2$ each identify $T(S^2 \setminus \{N\}) = \mathbb{R}^2 \times \mathbb{R}^2$ (and the same for $T(S^2 \setminus \{S\})$). [16] I'm not going to include the whole discussion on trivializations of S^2, so read §10 in Lee for this.

[14] chapter 8 lee, thank me later

[15] worst example of all time

[16] personal remark: trivializations is similar to the idea of parallelization (local vs global) pointed out in lee. a trivialization means the tangent bundle locally looks like the projection of $U \times \mathbb{R}^m$ onto U.

Given $f\colon M \to N$ smooth, define $Df\colon TM \to TN$, where $v \mapsto Df_p(v) \in T_{f(p)}N$ (for $v \in T_pM$).

Proposition 12.5. This is a smooth map.

Proof. Choose charts (U, φ) for M at p, (V, ψ) for N at $f(p)$. We get the commutative diagram,

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[r, "Df"] \arrow[d, "(\varphi\circ\pi, D\varphi)"] & \pi^{-1}(V) \arrow[d, "(\psi\circ\pi, D\psi)"] \\
\varphi(U) \times \mathbb{R}^m \arrow[r, "(\psi\circ f\circ\varphi^{-1}, D(\psi\circ f\circ\varphi^{-1}))"] & \psi(V) \times \mathbb{R}^n
\end{tikzcd}$$

Thus, Df is smooth by definition. $\square$

Consider, now, $\iota\colon M \hookrightarrow \mathbb{R}^n$ be a submanifold; we get a map $D\iota\colon TM \hookrightarrow \mathbb{R}^n \times \mathbb{R}^n$. This embeds TM as a submanifold into $\mathbb{R}^n \times \mathbb{R}^n$. We now discuss general vector bundles.

Definition 12.6. A manifold E^{m+n} equipped with

(i) A smooth map $\pi\colon E \to M^m$,

(ii) An n-dimensoinal vector space structure on each $\pi^{-1}(p)$,

(iii) For each $p \in M$, there's a neighborhood $U \ni p$ and a smooth map such that the diagram commutes and τ is linear on each $\pi^{-1}(p)$, i.e.,

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "\tau"] \arrow[dr, "\pi"'] & & \mathbb{R}^n \times U \arrow[dl, "\pi_2"] \\
& U &
\end{tikzcd}$$

The space E is usually called the total space of the vector bundle $\pi\colon E \to M$.

As some examples of vector bundles, we have

(i) The tangent bundle TM is a vector bundle.

(ii) The trivial bundle $\mathbb{R}^n \times M$.

(iii) The Möbius strip.

If we regard the Möbius strip as $\mathbb{R} \times S^1$, consider the charts $\mathbb{R} \times (0, 2\pi) \to S^1 \setminus \{1\}$ by $e^{i\theta} \circ \pi_2$ and $\mathbb{R} \times (-\pi, \pi) \to S^1 \setminus \{-1\}$ by $e^{i\theta} \circ \pi_2$ again. Then the transition maps between $\mathbb{R} \times (0, 2\pi) \to \mathbb{R} \times (-\pi, \pi)$ is given by $(\mathrm{id}, \mathrm{id})$ and the other direction by $(-\mathrm{id}, t \mapsto t - 2\pi)$.

Remark 12.7. Instead of $+\mathrm{id}$, we can use $(v, t) \mapsto (r(t)v, t)$, where r is any positive function to get the same structure.

In particular, M (the Möbius strip) is the tautological line bundle over $\mathbb{R}\mathrm{P}^1$ (i.e., every point of $\mathbb{R}\mathrm{P}^1$ is a line, so let the fiber be that line).

(iv) The tautological line bundle over $\mathbb{R}\mathrm{P}^n$; for each point, take the corresponding line; e.g., for $\mathbb{R}\mathrm{P}^2$, take the local trivialization

$$(xt, yt, t) \mapsto (t, (x : y : 1)),$$

from $\pi^{-1}(U)$ to $\mathbb{R} \times U$.

(v) The tautological k-plane bundle over $\mathrm{Gr}_k(\mathbb{R}^n)$.

Definition 12.8. A *smooth section* $s\colon M \to E$ of a vector bundle $\pi\colon E \to M$ is a smooth map such that $\pi \circ s = \mathrm{id}$. (For every $p \in M$, choose a vector above it).

Definition 12.9. A vector field on M is a smooth section of a tangent bundle.

We may check that this definition is commensurate to our definition from earlier in the lecture. We give some examples.

(i) There is a unique zero section sending $p \in M$ to $0 \in \pi^{-1}(p)$.

(ii) Sections of the trivial bundle $\mathbb{R}^n \times M$ (equivalently, maps $f\colon M \to \mathbb{R}^n$, identified by $f \leftrightarrow (f, \mathrm{id})$)

(iii) There's no nowhere zero section of the Möbius strip. This is because of the intermediate value theorem.

(iv) A *frame* is a smooth choice of basis at each point of M, i.e., a set of sections $s_1, \ldots, s_n\colon M \to E$ such that, for each p, $s_1(p), \ldots, s_n(p) \in E_p$ forms a basis in E_p. A global frame exists if and only if a trivialization of the bundle, i.e., a bundle map $F\colon E \to \mathbb{R}^n \times M$ by $(s_1, \ldots, s_n, \pi)$ commutes in the following daigram,

$$\begin{tikzcd} E \arrow[rr, "F := (s_1, \ldots, s_n, \pi)"] \arrow[dr, "\pi"'] & & \mathbb{R}^n \times M \arrow[dl, "\pi_2"] \\ & M & \end{tikzcd}$$

and F is linear on each fiber, giving an isomorphism on every fiber.

§13 Day 13: Vector Fields (Feb. 26, 2026)

We continue our discussion of vector bundles and tangent bundles. Note that we're working through §9 in the textbook.

Definition 13.1 (Global)**.** A vector field on M is a smooth section of the tangent bundle of M.

Definition 13.2 (Coordinate-wise)**.** A vector field is a way of assigning $p \in M$ to $X_p \in T_pM$ such that, for each chart (U, φ), $p \in U$ smoothly maps to $D\varphi(X_p)$.

For an example, consider $S^3 \subset \mathbb{R}^4$; for $p \in S^3$, obsevre that $T_pS^3 = \ker DF = \langle p \rangle^\perp$ for $F(x) = x \cdot x$. Indeed, this means

$$TS^3 = \{(p, v) \mid p \cdot p = 1, p \cdot v = 0\} \subset \mathbb{R}^4 \times \mathbb{R}^4,$$

where the condition $p \cdot p = 1$ can be regarded as $p \in S^3$, and $p \cdot v$ means $v \in T_pS^3$. In particular, if $\iota \colon S^3 \hookrightarrow \mathbb{R}^4$ is the inclusion, then we have an inclusion of tangent spaces $d\iota(TS^3) \subset T\mathbb{R}^4 \cong \mathbb{R}^4 \times \mathbb{R}^4$. For $p = (x_0, x_1, x_2, x_3) = (z, w) \in \mathbb{C}^2$, choose

$$\begin{aligned} v_1 &= (-x_1, x_0, -x_3, x_2) = (iz, iw), \\ v_2 &= (-x_2, -x_3, x_0, x_1) = (-w, z), \\ v_3 &= (x_3, -x_2, -x_1, x_0) = (-iw, iz). \end{aligned}$$

Notice that $p \mapsto (p, v_i)$ for $i = 1, 2, 3$ are vector fields (since $v_i(p) \perp p$, so $v_i(p) \in T_pS^3$), $v_i(p)$ for $i = 1, 2, 3$ is a unit vector for each $p \in S^3$, and $v_i \perp v_j$ (for $i \neq j$). This means $v_1(p), v_2(p), v_3(p)$ form a basis for each T_pS^3, and $\{v_1, v_2, v_3\}$ forms a *frame* for TS^3, meaning S^3 is parallelizable, and $TS^3 \cong \mathbb{R}^3 \times S^3$ (isomorphic as a bundle) under the map

$$(p, x) \mapsto (x_1, x_2, x_3),$$

where $x = x_1v_1(p) + x_2v_2(p) + x_3v_3(p)$. Notice, now, that v_1 is the velocity vector of a unit speed curve that traces out the fibers of the Hopf fibration.

We present a third definition of vector fields; to start, though, recall that vectors in T_pM are directional derivatives $v \colon C^\infty(M) \to \mathbb{R}$ defined by

$$v(f) = \frac{d}{dt}\bigg|_{t=0} (f \circ \gamma_v),$$

where $\gamma_v \colon (a, b) \to M$ is some curve with $\gamma_v(0) = p$.

Definition 13.3. A function $p \in M$ to $X_p \in T_pM$ is a vector field if, for every $f \in C^\infty(M)$, $p \mapsto X_p(f)$ ("the directional derivative of f at p") is a smooth function $M \to \mathbb{R}$.

Proposition 13.4. The coordinate-wise definition is equivalent to the third definition.

Proof. Let X be a smooth function for all $f \in C^\infty(M)$, which holds if and only if, for all $f \in C^\infty M$, there are charts (U, φ) such that

$$\begin{tikzcd} U \arrow[r, "X(f)"] \arrow[d, "\varphi"] & \mathbb{R} \\ \varphi(U) \arrow[ur, "X(f)\circ\varphi^{-1}"'] & \end{tikzcd}$$

the composition above is smooth. This is equivalent to $D_{D\varphi(X_p)}(f \circ \varphi^{-1})$ is a smooth function (where D here is the directional derivative), which is equivalent to there being charts (U, φ) such that $D\varphi_p(X_p)$ is a smooth function $U \to \mathbb{R}^m$.[17] $\square$

Definition 13.5. A vector field is a linear map $C^\infty(M) \to C^\infty(M)$ satisfying $X(fy) = X(f)g + fX(g)$.

A linear map of this form in an algebra is called a *derivation*.

(i) X_p is defined by $X_p(t) = [X(f)](p)$.

(ii) X_p satisfies the product rule $X_p(fg) = X_p(f)g(p) + f(p)X_p(g)$.

This means definition 4 implies definition 3, which is left as an exercise. Also,

(iii) You can multiply vector fields by smooth functions, so the vector fields on M form a module over $C^\infty(M)$.

[17] fedya continued onwards with a really weird direction for this proof.

§14 Day 14: Lie Brackets and Flow (Mar. 3, 2026)

Recall that a vector field on M can be regarded as a section of the tangent bundle $X\colon M \to TM$ such that $\pi \circ X = \mathrm{id}_M$, or a linear map $X\colon C^\infty(M) \to C^\infty(M)$ satisfying the proudct rule

$$X(fg) = X(f)g + fX(g).$$

Note that $X_p \in T_pM$ is defined by $X_p(f) = X(f)(p)$, i.e., a directional derivative, and is therefore a vector.

Definition 14.1. We denote $\mathfrak{X}(M)$ to be the collection of vector fields on M.

In general, the composition of two vector fields $X, Y\colon C^\infty(M) \to C^\infty(M)$ is not necessarily a vector field[18]; indeed, if $M = \mathbb{R}^n$, $X = Y = \partial_x$, then this clearly does not work, since

$$(X \circ Y)(fg) = X(Y(f)g + fY(g)) \implies \partial_{xx}(fg) = \partial_x(f_x g + f g_x) = f_{xx}g + 2f_x g_x + f g_{xx}.$$

However, the *Lie bracket*

$$[X, Y] := X \circ Y - Y \circ X$$

is always a vector field. In general, commutators of derivations are derivations. The Lie bracket satisfies $[Y, X] = -[X, Y]$, and may in general be written, for

$$X = \sum_{i=1}^m a^i \frac{\partial}{\partial x^i}, \quad Y = \sum_{i=1}^m b^i \frac{\partial}{\partial x^i},$$

as

$$X \circ Y = \sum_{i,j=1}^{,} a^i \frac{\partial b^i}{\partial x^j}\frac{\partial}{\partial x^i} + \sum_{i,j=1}^m a^i b^j \frac{\partial^2}{\partial x^i \partial x^j} \implies [X, Y] = \sum_{i,j=1}^m \left(a^j \frac{\partial b^i}{\partial x^j} - b^j \frac{\partial a^i}{\partial x^j}\right)\frac{\partial}{\partial x^i}.$$

Note that this calculation applies to general manifolds by taking local coordinates.[19] We now give some examples of computations.

(i) Let $X = \partial_i$, $Y = \partial_j$, with coefficients a^i, b^i constant functions. Then $\partial_j a^i = 0$, so the commutator is zero. The definition is coordinate-free, so if you can take a chart where your vector fields are coordinate vector fields, then $[X, Y] = 0$. One such example, is given by considering partial derivatives with respect to latitude and longitude on S^2 away from the poles.

(ii) In $\mathbb{R}^3$, let

$$X = \partial_x, \quad Y = \partial_y + x\partial_z.$$

Then $[X, Y] = \partial_x(x)\partial_z = \partial_z$. The geometric interpretation is, how does going in the direction of X then Y differ form going in the direction of Y then X?

(iii) We have $[X, fY](g) = X \circ (fY)(g) - (fY) \circ X(h)$; after algebra,

$$[X, fY] = X(f)Y + f[X, Y].$$

(iv) The *Jacobi identity*:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

This is also true for derivations.

[18] since chain rule and product rule are not necessarily commensurate

[19] see marco gualteri's here, p.88-89

We now give more definitions. A vector space over a field of characteristic $\neq 2$ with an operation $[\bullet, \bullet]$ which is bilinear, anticommutative, and satisfies the Jacobi identity is called a *Lie algebra*. An example is $\mathfrak{X}(M)$. Indeed, we say $X \in \mathfrak{X}(M)$ is *tangent* to a submanifold S if, for every $p \in S$,

$$X_p \in T_pS \subset T_pM,$$

so X restricted to S is a vector field on S. If X, Y are tangent to S, then so is their commutator. We may prove this by considering that X is tangent to S if and only if, for every $p \in S$, $X_p(f)$ depends only on the restriction of f to S.

We now discuss the flow of a vector field. The idea is, "a vector field can be used to describe ocean currents". We want a function $\phi(t, x)$ describing where a particle starting at x at time 0 ends up at time t.

Definition 14.2. Let $X \in \mathfrak{X}(M)$. A curve $\gamma\colon (a, b) \to M$ is a *solution curve* for X if $\gamma'(t) = X_{\gamma(t)}$ for all $t \in (a, b)$.

Indeed, we have existence and uniqueness just like we saw in MAT267. Consider $M = U \subset \mathbb{R}^m$; let $\gamma(t) = x(t) = (x^1(t), \ldots, x^m(t))$. Then γ is a solution curve if it satisfies the first-order ODE $x'(t) = a(x(t))$. As an example, for some $a = (a^i)_{i=1}^m$ constant,

$$X = \sum_{i=1}^m a^i \frac{\partial}{\partial x^i} \implies \gamma_p(t) = p + ta.$$

Theorem 14.3 (Existence and Uniqueness)**.** Let $U \subset \mathbb{R}^m$ be open and $a \in C^\infty(U, \mathbb{R}^m)$. For every $p \in U$, there exists an open interval $J_p \subset \mathbb{R}$ containing 0 and a unique solution $x_p\colon J_p \to U$ such that $x_p'(t) = a(x_p(t))$, $x_p(0) = p$, and J_p is maximal.

Indeed, define

$$\phi(t, p) = x_p(t), \quad \frac{\partial}{\partial t}\phi(t, p) = a(\phi(t, p)).$$

The maximal domain of definition of ϕ is an open subset $J \subset \mathbb{R} \times U$ containing $\{0\} \times U$, and ϕ is smooth on J. On a manifold, for $X \in \mathfrak{X}(M)$ and each $p \in M$, there exists a maximal interval J_p and a unique solution $\gamma_p\colon J_p \to M$ such that

$$\gamma_p'(t) = X_{\gamma_p(t)}, \quad \gamma_p(0) = p.$$

The map $\phi(t, p) = \gamma_p(t)$ is smooth and its maximal domain is open.

Proof sketch. If solutions exist and are unique in charts, then we have local existence and uniqueness. Since the maximal interval is open, if an endpoint existed, local existence would extend the solution; uniqueness followed by extending past the largest point where two solutions agree, and smoothness follows from the version of flow in $\mathbb{R}^m$, along with the *semigroup* or *flow* property

$$\phi_t \circ \phi_s(p) = \phi_{t+s}(p),$$

whenever both sides are defined. $\square$

Corollary 14.4. For fixed t, $\phi_t\colon U \to \phi_t(U)$ is a diffeomorphism in M.

§15 Day 15: Flow (Mar. 5, 2026)

We recap last lecture; let X be a vector field on M; then X yields $\phi\colon \mathscr{J} \to M$, where U is an open subset of $\mathbb{R} \times M$ containing $\{0\} \times M$. We denote $\phi_t(p)$ to be the flow of p follownig X at time t. Formally, $\phi_t(p) = X_{\phi_t(p)}$ is smooth and unique where defined. We now discuss the flow property. If $\phi_t(p)$ is defined, then $\phi_s(\phi_t(p)) = \phi_{s+t}(p)$.

Corollary 15.1. ϕ_t is a diffeomorphism onto its image, $\phi_t\colon \mathscr{J} \cap (\{t\} \times M) \to \phi_t(U)$, where we denote the domain as $\{t\} \times U$ for convenience.

Proof. Consider $s = -t$. □

Definition 15.2. A vector field X is complete if $\mathscr{J} = \mathbb{R} \times M$ (i.e., flow exists at all time, at every point)

As an example (§6.29), $x\partial_x$ has flow $\phi_t(x) = e^{tx}$, so it is a complete vector field. $x^2\partial_x$ does not yield a complete vector field because it escapes to infinity in finite time.

Theorem 15.3 (§6.32)**.** If X has compact support, i.e., it is 0 outside a compact set, then its complete.

Corollary 15.4. If X is a vector field on a compact manifold, then its complete.

Proof. Suppose $p \notin \operatorname{supp} X$, so $X_p = 0$; then $\phi_t(p) = p$ for all $t \in \mathbb{R}$. This means $p \in \operatorname{supp} X$ implies $\phi_t(p) \in \operatorname{supp}(X)$ wherever defined. Recall that the domain $\mathscr{J}$ of ϕ is open. Then, for all $\varepsilon > 0$, we have an open subset $(-\varepsilon, \varepsilon) \times U_\varepsilon$, where

$$U_\varepsilon := \{p \in M : \phi_t(p) \text{ is defined for } t \in (-\varepsilon, \varepsilon)\}.$$

Clearly, such U_ε covers M, so there exists some finite subcover $\{U_{\varepsilon_i}\}$, and we may consider that $U_{\varepsilon_{\min}} = M$. We now apply the flow property;

$$\phi_{\varepsilon_{\min/2}}(p)$$

is defined for all $p \in M$, so by the flow property, $\phi_{n\varepsilon_{\min}}(p)$ is defined for all $p \in M$ and $n \in N$, so $\phi_t(p)$ is defined for all $t \geq 0$. We may handle $t < 0$ the exact same way. □

Overall, this means that if X is a complete vector field, then its flow defines a 1-parameter family of diffeomorphisms ϕ_t for $t \in \mathbb{R}$. In particular, flow satisfies the semigroup property (w.r.t. composition) and $\phi(t, p)$ is a smooth function on $M \times \mathbb{R}$. Conversely, if ϕ_t satisfies $\phi_t \circ \phi_s = \phi_{t+s}$ and is smooth on $M \times \mathbb{R}$, then its the flow of a vector field X, i.e.,

$$X_p(f) = \frac{d}{dt}\bigg|_{t=0} f(\phi_t(p)).$$

In other words, X_p is the velocity vector in the sense $\gamma(t) = \phi_t(p)$.

Proof. First, X_p satisfies the product rule, i.e.,

$$X_p(fg) = \frac{d}{dt}\bigg|_{t=0} f(\phi_t(p))g(\phi_t(p)),$$

so it is a vector field. Second, $\phi_t(p)$ is an integral curve of X because

$$\phi_t(p) = \frac{d}{ds}\bigg|_{s=0} \phi_s(\phi_t(p)) = X_{\phi_t(p)}.$$ □

As an example, given $A \in M_{m\times m}(\mathbb{R})$, let $\phi_t \colon \mathbb{R}^m \to \mathbb{R}^m$, where

$$\phi_t(x) = e^{tA}x = \left(\sum_{N=0}^{\infty} \frac{t^N}{N!} A^N x\right).$$

The corresponding vector field is

$$X(f)(x) = \frac{d}{dt}\bigg|_{t=0} f(e^{tA}x) = \nabla f \cdot \frac{d}{dt}\bigg|_{t=0} (e^{tA}x) = \nabla f \cdot Ax,$$

so $X = \nabla \cdot Ax$.

Definition 15.5 (Pullback)**.** Given $F \colon M \to N$, we define F^*f, for $f \colon N \to \mathbb{R}$, as just $f \circ F \colon M \to \mathbb{R}$. If F is a diffeomorphism, then we may pull back vector fields, i.e., F^*X is given by $X_p = DF(F^*X_{F^{-1}(p)})$.

There is another set of definitions for Lie derivative $\mathcal{L}_X f = \frac{d}{dt}\big|_{t=0} \phi_t^* f \in C^\infty(M)$, where X is a vector field on M We may check that $\mathcal{L}_X f = X(f)$.

§16 Day 16: Lie Derivatives, Frobenius (Mar. 10, 2026)

Recall last class that we defined the *Lie derivative* of a vector field (§6.39),

$$\mathcal{L}_X Y = \frac{d}{dt}\bigg|_{t=0} \Phi_t^* Y \in \mathfrak{X}(M),$$

where Φ_t denotes the flow of X, which we know is a diffeomorhpism. In particular, it also induces the isomorphism $d(\Phi_t)_p \colon T_pM \to T_{\Phi_t(p)}M$, which gives us a way to compare vectors in these spaces.

Theorem 16.1 (§6.40)**.** If Y is also a vector field, then $\mathcal{L}_X Y = [X, Y]$.

Proof. Given $f \in C^\infty(M)$, we want to show that $\mathcal{L}_X Y(f) = [X, Y](f)$. Indeed, we have, by definition of pullback,

$$\Phi_t^*(Y(f)) = (\Phi_t^* Y)(\Phi_t^* f).$$

By taking the derivative with respect to t at $t = 0$ of both sides, we get[20]

$$X(Y(f)) = \left(\frac{d}{dt}\bigg|_{t=0} \Phi_t^* Y\right)(f) + Y\left(\frac{d}{dt}\bigg|_{t=0} \Phi_t^* f\right) = (\mathcal{L}_X Y)(f) + Y(X(f)),$$

where, by rearranging, we get $\mathcal{L}_X Y = [X, Y]$ as desired. □

The upshot is that $[X, Y]$ intuitively describes "how X changes along the flow of Y".

Theorem 16.2 (§6.41)**.** Let Φ be the flow of X, and let Ψ be the flow of Y. We have $[X, Y] = 0$ if and only if $\Phi_t^* Y = Y$, or, equivalently, $\Phi_t^* X = X$. This is actually also equiaveltnt to $\Phi_t \circ \Psi_s = \Psi_s \circ \Phi_t$.

As an example, consider $X = \partial_x$ on $\mathbb{R}^2$; then $[X, Y] = 0$ if and only if Y is invariatn under shifts in the x direction, such as $Y = \partial_x + y\partial_y$. We may visualize this by drawing flow lines.

Proof. We want to show that $[X, Y] = 0 \iff \Phi_t^* Y = Y \iff \Psi_t^* X = X$ is equivalent to $\Phi_t \circ \Psi_s = \Psi_s \circ \Phi_t$. In the forwards direction, if $\Phi_t^* Y = Y$, then Φ_t takes integral curves of Y to integral curves of Y. In particular, this means $\Phi_t^* \Psi_s = \Psi_s$, in the sense that $\Phi_t \circ \Psi_s \circ \Phi_t^{-1} = \Psi_s$.[21] □

More generally,[22]

$$[X, Y] = \lim_{t \to 0} \frac{1}{t^2} \left(\Phi_t^* \Psi_t^* - \Psi_t^* \Phi_t^*\right).$$

As an example, in $\mathbb{R}^3$, consider $X = \partial_x$, $Y = \partial_y + x\partial_z$. We see $[X, Y] = \partial_z$, but we can also see it by seeing

$$\frac{1}{st}\left(\Psi_s^* \Phi_t^* - \Phi_t^* \Psi_s^*\right) \to \frac{\partial}{\partial z}.$$

Last lecture, someone asked about what the Jacobi's identity was useful for. So we digress here! Recall Jacobi's identity,

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

[20] here, fedya did some expanding that made no sense and was ultimately pointless. just read the book if you're confused here, yeah? oh and he also made an error.

[21] this is not a complete proof, so... read the textbook!

[22] oh no

It can be viewed as a product rule where $[\bullet, \bullet]$ is both the derivation an product by rewriting (by using anticommutativity) as

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]] \iff \mathcal{L}_X[Y, Z] = [\mathcal{L}_X Y, Z] + [Y, \mathcal{L}_X Z].$$

We now introduce the Frobeinus theorem. When can you take linearly independent vector fields $X_1, \ldots, X_k$ and "integrate them" to get a k-dimensional submanifold, i.e., for every $p \in M$, there is locally a k-dimensional submanifold through p such that $X_1, \ldots, X_k$ are tangent to S near p?

Theorem 16.3 (Frobenius)**.** The above happens if and only if $[X_i, X_j] = \sum_{j=1}^{r} c_{ij}^k(p) X_k$ for all i, j. This is called the *Frobenius condition.*

As an example, consider $M = T^2$ (2-torus), and $X_1 = 2\partial_x + 3\partial_y$. The flow lines, when drawn on the gluing diagram for the 2-torus, form "foliations by submanifolds" (cf. p.151-153). As another example, consider $X = \partial_x + \sqrt{2}\partial_y$; then the flow lines $\mathbb{R} \to T^2$ define a *foliation* (i.e., locally nice-looking partitions into submaifolds), but they are not globally manifolds.

Lemma 16.4 (Flow Straightening Lemma, §6.47)**.** Let $X \in \mathfrak{X}(M)$ be a vector field, and $p \in M$ such that $X_p \neq 0$. There there exists a coordinate chart (U, φ) about p, with corresponding local coordinates $x^1, \ldots, x^m$ such that

$$X\big|_U = \frac{\partial}{\partial x^1}.$$

Proof. Choose a submanifold N of codimension 1 such that $X_p \notin T_p N$. The idea is that we will use N as the "$x_1 = 0$" points along X to get the extra coordinate. Consider $\Phi\colon (-\varepsilon, \varepsilon) \times N \to M$; then $D_{(0,p)}\Phi(s, v) = v + sX_p$, so Φ has full rank at $(0, p)$, and so, by the inverse function theorem, it is a chart for M near p. By definition, X_p is ∂_1 with respect to these coordinates.[23] $\square$

We are now going to upgrade this lemma[24].

Lemma 16.5. Let $p \in M$, and let $X_1, \ldots, X_r \in \mathfrak{X}(M)$ be vector fields, whose values at $p \in M$ are linearly independent with

$$[X_i, X_j] = 0$$

for all $1 \leq i, j \leq r$. Then there exists a coordinate chart (U, φ) near p, with corresponding local coordinates $x^1, \ldots, x^m$ such that

$$X_1\big|_U = \frac{\partial}{\partial x^1}, \ldots, X_r\big|_U = \frac{\partial}{\partial x^r}.$$

Proof. We run through the previous proof with N a codimension r submanifold. By taking

$$F\colon (-\varepsilon, \varepsilon)^r \times N \to M;$$

then $F(t_1, \ldots, t_r, p) = \Phi_{t_1}^{X_1} \circ \cdots \circ \Phi_{t_r}^{X_r}(p)$, where we may rearrange the flows however we like because the vector fields commute. Thus,

$$DF_{(0,\ldots,0,p)}\Phi(s_1, \ldots, s_r, v) = v + s_1 X_1\big|_p + \cdots + s_r X_r\big|_p. \qquad \square$$

[23] read the book p.149 fedya legit said this is a froof

[24] fedya's words not mine

We now prove the Frobenius theorem.[25]

Proof. The Frobenius condition is necessary; it is true because, if X_i, X_j are tangent to a submanifold S, then so is $[X_i, X_j]$. The Frobenius condition is also sufficient beacuse we may reduce to the $[\cdot,\cdot] = 0$ case, i.e., solve it in $U \subset \mathbb{R}^m$, in coordinates

$$X_i = \sum_{j=1}^{r} a_{ij}(\vec{x}) \frac{\partial}{\partial x^j} + \sum_{j=r+1}^{m} b_{ij}(\vec{x}) \frac{\partial}{\partial x^j},$$

where $a_{ij}(\vec{0}) = I_r$ and $b_{ij}(\vec{0}) = 0$, which is done by a linear change of coordinates. Define $X_i = \sum_{j=1}^{r} a_{ij} X'_j$. They span the same thing and commute, so by the flow straightening lemma, they give a submanifold. $\square$

[25] read the book don't read these notes. ive said it like 10 times today now

§17 Day 17: Differential Forms (Mar. 12, 2026)

Let us summarize what we've done with vector fields. To understand vector fields, we need to understand them:

(i) Coordinate-wise: $X = \sum_i X^i \partial_{x_i}$, in any chart (U, φ) on M.

(ii) Abstractly: as a smooth section of the tangent bundle, i.e., a smoort choice of tangent vector at each point.

(iii) Functional: a derivation $C^\infty(M) \to C^\infty(M)$.

(iv) Integral: something that "generates" a flow, i.e., a 1-parameter family of diffeomorphisms.

We move from vector fields onto differential forms now (i.e., §7 in Meinrenken).[26] We can discuss the previous 4 ways to define a vector field for differential forms as well.

(i) Coordinate-wise: a k-form ω is a sum

$$\sum_{|I|=k} f_I \, d\omega_{i_1} \wedge \cdots \wedge d\omega_{i_k},$$

where I is a collection of k indices $i_1, \ldots, i_k$.

(ii) Abstract: ω is a smooth section of a k-covector at each point, i.e., a section of some vector bundle.

(iii) Functional: a form is a kind of function

$$\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M),$$

where we consider the product to be a k-tuple.

(iv) Integral: a k-form is something you can integrate over a k-dimensional submanifold.

We give an example. Given $f \in C^\infty(M)$, define the (exterior) differential $df \colon \mathfrak{X}(M) \to C^\infty(M)$, where $df(X) = X(f)$. Note that df is $C^\infty(M)$-linear (i.e., a linear map in functions). Specifically,

$$df(gX + hY) = g\, df(x) + h\, df(y).$$

This is left as an exercise. We now expand on the functional definition.

Definition 17.1 (§7.7)**.** A 1-*form* on M is a $C^\infty(M)$-linear map from $\mathfrak{X}(M)$ to $C^\infty(M)$.

The set of 1-forms is denoted $\Omega^1(M)$. In particular, $\Omega^1(M)$ can be regarded as a module over the algebra $C^\infty(M)$.[27]

Definition 17.2 (p.159; §7.2)**.** Given a vector space V, we define its *dual* $V^* = \mathrm{Hom}(V, \mathbb{R})$.

Given $\alpha \in V^*$, we can write $\alpha(v)$ as $\langle \alpha, v \rangle$ to emphasize that V is finite-dimensional. Specifically, $(V^*)^* = V$ canonically. We can think of v as a homomorphism from $V^* \to \mathbb{R}$. Indeed, in this case, we also have $V^* \cong V$, but not canonically unless we first fix an inner product.

[26] "i'm now going to say a bunch of words, and i'm not sure how much it will mean to you" aka welcome back to DEFINITIONS!!

[27] see §7.9 ish, p.164-165

Definition 17.3. Given $e_1, \dots, e_n$ a basis for V, we get $e^1, \dots, e^n$ the *dual basis* for V^*, defined by $e^i(e_j) = \delta_{ij}$ (i.e., Kronecker delta).

Given $p \in M$, we have the tangent space T_pM. Its dual is the tangent space T_p^*M, where elements $v \in T_p^*M$ are called *covectors*. In $T_p\mathbb{R}^n$, we have a distinguished basis $\partial_1, \dots, \partial_n$; its dual basis is written $dx_1, \dots, dx_n$, where

$$dx_i(a_1\partial_1 + \cdots + a_n\partial_n) = a_i.$$

Definition 17.4. Given a linear map $T\colon V \to W$, the dual map goes the other way, i.e., $T^*\colon W^* \to V^*$, defined by $T^*\omega(v) = \omega(Tv)$.

As an example, given a smooth map $f\colon M \to N$, we have $Df_p\colon T_pM \to T_{f(p)}N$ with $v \mapsto f_*v = Df_p(v)$. The dual map takes $Df_p^*\colon T_{f(p)}^*N \to T_p^*M$ by $v \mapsto f^*v = (Df_p)^*(v)$. See §7.4.

As another example, given $f \in C^\infty(M)$, we can define a covector $df_p \in T_p^*M$ by $df_p(v) = v(f) = D_vf$. In $\mathbb{R}^n$, we can write

$$df_p = \frac{\partial f}{\partial x_1}dx_1 + \cdots + \frac{\partial f}{\partial x_n}dx_n.$$

This is also left as an exercise. Recall that, given $f\colon M \to N$ and $g\colon N \to \mathbb{R}$¡ we define $f^*g = g \circ f\colon M \to \mathbb{R}$.

Proposition 17.5 (§7.6)**.** Given $f\colon M \to N$ smooth and $g\colon N \to \mathbb{R}$ smooth, we have $d(f^*g)_p = f^*dg_{f(p)}$.

Proof. This is an exercise in chasing definitions. For every $v \in T_pM$, we have

$$\langle d(f^*g)_p, v\rangle = v(f^*g) = v(g \circ f) = Df_p(v)(g) = \left\langle (dg)_{f(p)}, T_pf(v)\right\rangle = \left\langle f^*dg_{f(p)}, v\right\rangle. \quad \square$$

§18 Day 18: One-forms (Mar. 17, 2026)

A 1-form is a smooth choice of covectors at each point on a manifold M.

Definition 18.1 (Coordinate-wise)**.** A map sending $p \mapsto \alpha_p$ is a one-form if and only if, for every chart (U, φ) on M,

$$\alpha_{\varphi(p)} = \sum_{i=1}^{m} g_i(\varphi(p))\, dx_i,$$

where the $g_i \colon \varphi(U) \to \mathbb{R}$ are smooth functions.

Consider the following examples,

(i) Consider $f \colon M \to \mathbb{R}$; then on a chart, we have

$$df = \sum_{i=1}^{m} \frac{\partial f}{\partial x_i} dx_i.$$

We leave it as an exercise to show that this is independent of choice of chart.

(ii) $y\, dx$ is a form on $\mathbb{R}^2$, and is not df for any $f \in \mathbb{R}^2 \to \mathbb{R}^2$. This is because, if such an f did exist, we would have

$$0 = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 1.$$

Definition 18.2 (Functional)**.** A 1-form is a $C^\infty(M)$ linear map $\mathfrak{X}(M) \to C^\infty(M)$ such that, at each point $v \in M$, we apply a covector $\alpha(v)$.

Proposition 18.3. Let α be a one-form on M; then there is a unique covector α_p, $p \in M$, such that

$$[\alpha(X)](p) = \alpha_p(X_p), \forall X \in \mathfrak{X}(M).$$

Proof. We want to show that if $X_p = Y_p$, then $\alpha(X)(p) = \alpha(Y)(p)$, then this gives a well-defined map $X_p \mapsto \alpha_p(X_p)$. Then, we need to show that it is linear, which will make it a covector. By linearity, it is enough to show that if $X_p = 0$, then $\alpha(X)(p) = 0$; if $X_p = 0$, we can write

$$X = \sum_{i=1}^{m} f_i y_i,$$

where $f_i(p) = 0$; since on some chart, we can have $y_i = \partial_i$ and then extend somehow, we have

$$\alpha(X)(p) = \sum_i \underbrace{f_i(p)}_{=0} \alpha(y_i)(p) = 0. \qquad \square$$

Proposition 18.4 (§7.10)**.** Similarly to vector fields, one-forms on M can be restricted to open subsets $U \subset M$. The two definitions given above are equivalent.

Recall that the pullback of a function $g \colon N \to \mathbb{R}$ along $f \colon M \to N$ is defined as $f^*g = g \circ f$, so we have that

$$Df \colon T_pM \to T_{f(p)}N, \quad v \mapsto f_* v.$$

This induces $T^*_{f(p)}N \to T^*_pM$ by $\nu \mapsto f^*\nu$. Now, given a one-form $\omega \in \Omega^1(N)$, we define its pullback by $(f^*\omega)_p = f^*(\omega_{f(p)})$.

Proposition 18.5 (§7.12)**.** This is again a one-form.

Proof. We use the coordinate-wise definition. We have a smooth function

$$f\colon (U \subset \mathbb{R}^m) \to (V \subset \mathbb{R}^n), \quad \omega = \sum_{i=1}^{n} g_i \, dx_i \in \Omega^1(V).$$

We have that Df_p has smooth coordinates for $\partial_{x_j} f_i$, so, notice that the restriction of f^* onto $f(p)$ is actually $(Df_p)^\top$. This means

$$(f^*\omega)_p = f^*\big|_{f(p)}(\omega_{f(p)}) = f^*\big|_{f(p)} \sum_i g_i(f(p)) \, dx_i,$$

which is a composition of smooth maps, whence it is a one-form. □

The pullback enjoys a few properties.

(i) $(f \circ g)^* = g * f^*$,

(ii) If $g\colon M \to \mathbb{R}$ is a smooth function, then $f^*(g\omega) = f^*g \cdot f^*\omega$,

(iii) $f^*(dg) = d(f^*g)$,

(iv) For some coordinates, we have that $(f^*\omega)_p$ is given by

$$f^*\big|_{f(p)} \sum_{i=1}^{n} g_i(f(p)) \, dx_i = \sum_{i=1}^{n} g_i(f(p)) f^*\big|_{f(p)} \, dx_i = \sum_{i=1}^{n} \sum_{j=1}^{m} g_i(f(p)) \frac{\partial f_i}{\partial x_j} \, dx_j.$$

Definition 18.6 (Abstract)**.** A one-form is a section of the cotangent bundle, i.e., for a smooth manifold M, the disjoint union

$$T^*M = \bigsqcup_{p \in M} T_p^*M$$

is called the cotangent bundle of M. It has a natural projection map $\pi\colon T^*M \to M$ such that $\omega \in T_p^*M$ is sent to p.

Note that, with tangent bundles, $f\colon M \to N$ defined $DF\colon TM \to TN$; we can only do this sort of thing when f is a diffeomorphism. On the other hand, the pullback of forms is given by: if $\omega \in \Omega(N)$ is defined as $s_\omega\colon N \to T^*N$, then $f^*\omega$ is given by

$$s_{f^*\omega}(p) = f^*\big|_{f(p)}(s_\omega \circ f(p)).$$

We now discuss integration of 1-forms along a curve (§7.6 in book). Given a smooth curve $\gamma\colon I \to M$ and $\alpha \in \Omega^1(M)$, we can take $\gamma^*\alpha \in \Omega^1(I)$, with

$$\gamma^*\alpha = f(t) \, dt$$

for some function $f(t)$. We define

$$\int_\gamma \alpha = \int_a^b f(t) \, dt,$$

where $I = [a, b]$. By the fundamental theorem of calculus, we get the 1-dimensional version of Stokes' theorem.

Theorem 18.7 (Stokes, 1D; §7.15)**.** Let $\alpha = df$ for some $f\colon M \to \mathbb{R}$. Then

$$\int_\gamma \alpha = f(\gamma(b)) - f(\gamma(a)).$$

Proof. The proof is immediate.

$$\gamma^* df = d(\gamma^a stf) = d(f \circ \gamma) = \frac{\partial(f \circ \gamma)}{\partial dt}\, dt,$$

then we may conclude by FTC. □

We also have that the integral is invariant with respect to reparameterizations. Indeed, we want to show that $\int_\gamma \alpha$ only depends on the curve as a submanifold of M, and the form α. Formally,

$$\int_\gamma \alpha = \pm \int_{\gamma\circ\kappa} \alpha,$$

where $\kappa\colon I \to I$ is a diffeomorphism and $\pm$ depends on the sign of $\frac{d\kappa}{dt}$.

Proof. We may write,

$$\int_{\gamma\circ\kappa} \alpha = \int_c^d (\gamma \circ \kappa)^* \alpha = \int_c^d \kappa^* \gamma^* \alpha = \int_{\kappa(c)}^{\kappa(d)} \gamma^* \alpha = \pm \int_a^b \gamma^* \alpha.$$ □

Recall that α is an *exact* form if it is the exterior derivative of some function $f\colon M \to \mathbb{R}$. In this case, $\int_\gamma \alpha$ depends only on the endpoints. In fact, this is an if and only if statement; if $\int_\gamma \alpha$ depends only on the endpoints, then define

$$f(p) = \int_\gamma \alpha,$$

where γ is any curve from p_0 to p. As it turns out, f is smooth and $df = \alpha$. As an example, let $M = S^1$ and $\alpha = d\theta$. Then

$$\int_{S^1} d\theta = \int_\gamma d\theta = 2\pi,$$

where $\gamma\colon [a, b] \to S^1$. Since we are integrating over a loop, and we get something nonzero, $d\theta$ cannot be an exact form.

§19 Day 19: k-forms (Mar. 19, 2026)

Last time, we showed that if α is a one-form, then

$$\int_\gamma \alpha = \int_a^b \gamma^* \alpha = \int_a^b \alpha_{\gamma(t)} \gamma_*(\partial_t))\, dt,$$

where $\gamma_*(\partial_t) \in T_{\gamma(t)}M$. Notice that this doesn't depend on the parametrization of γ; this is because $\alpha_{\gamma(t)}(kv) = k\alpha_{\gamma(t)}(v)$, i.e., $\int$ remains the same by substitution. Moreover, we have that $\alpha(X+Y) = \alpha(X) + \alpha(Y)$, which doesn't matter for this computation. Now, we want a k-form to be a thing that can be integrated over a k-dimensinoal submanifold. Formally, this is a map $\omega\colon T_pM \times \cdots \times T_pM \to \mathbb{R}$, where we can define, for $F\colon U \to M$,

$$\int_{F(U)} \omega = \int_U \omega_{F(x)}(F_*\partial_1, \dots, F_*\partial_k)\, dx_1 \dots dx_k.$$

Once again, we have invariance under reparametrization by diffeomorphisms $G\colon U \to V$. Indeed, we want

$$\begin{aligned}\int_{G(U)} &\omega_{F(G^{-1}(y))} \left((F \circ G^{-1})_x \partial_1, \dots, (F \circ G^{-1})_x \partial_k\right)\, dy \\ &= \pm \int_U \omega_{F(x)} \left((F \circ G^{-1})_x \partial_1, \dots, (F \circ G^{-1})_x \partial_k\right) \left|\det DG^{-1}\right| dx,\end{aligned}$$

where $\pm$ is there in case G changes orientation. Therefore, we need:

Fact 19.1. Let $\omega\colon V \times \cdots \times V \to \mathbb{R}$ be a map which is multilinear, i.e., for all $i = 1, \dots, k$, if $v_k = au + bw$, then

$$\omega(v_1, \dots, v_k) = a\omega(v_1, \dots, u_i, \dots, v_k) + b\omega(v_1, \dots, w_i, \dots, v_k),$$

and skew-symmetric, i.e., alternating, where

$$\omega(v_1, \dots, v_k) = -\omega(v_1, \dots, v_i, v_{i-1}, \dots, v_k);$$

then $\omega(v_1, \dots, v_k) = C \cdot \det[v_1 \mid \cdots \mid v_k]$.

Therefore, if $\omega\colon V \times \cdots \times V \to \mathbb{R}$ is multilinear and alternating, it is called a *alternating k-tensor*, where its restriction to any k-dimensional subspace behaves like a determinant.

Definition 19.2 (Vague)**.** A k-form $\omega \in Omega^k(M)$ is a smooth choice of alternating k-tensors $\omega_p\colon T_pM \times \cdots \times T_pM \to \mathbb{R}$ at each $p \in M$.

Definition 19.3 (Formal, §7.20)**.** A k-form is a map $\omega\colon \mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M)$ which is $C^\infty(M)$-linear in each coordinate, and (i) skew-symmetric.

A n-form $\omega \in \Omega^n(\mathbb{R}^n)$ always takes the form $\omega(X_1, \dots, X_n)(p) = f(p) \det[(X_1)_p \mid \cdots \mid (X_n)_p]$, where $f\colon \mathbb{R}^n \to \mathbb{R}$ is some smooth function. What do alternating k-tensors on $\mathbb{R}^n$ look like in general?

Definition 19.4. Let $I = \{i_1 < \cdots < i_k\} \subset \{1, \dots, n\}$[28] Define $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ to be the alternating k-tensors, which is, $dx_I(\partial_{i_1}, \dots, \partial_{i_k}) = 1$, $dx_I(\partial_j, \dots) = 0$ if $j \notin I$. This uniquely defines an alternating k-tensor.

[28] for fucks sake bestie just say k indices.

We may interpret the above as, taking $v_1, \ldots, v_k$ and computing $\det(\pi_I v_1, \ldots, \pi_I v_k)$, where $\pi_I \colon \mathbb{R}^n \to \langle \partial_{i_1}, \ldots, \partial_{i_k} \rangle$.

Lemma 19.5. Every alternating k-tensor $\omega \colon \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ can be epxressed uniquely as

$$\omega = \sum_{|I|=k} \omega_I \, dx_I, \quad \omega_I \in \mathbb{R}.$$

Proof. Let $\omega_I = \omega(\partial_{i_1}, \ldots, \partial_{i_k})$. Check that $\omega(v_1, \ldots, v_k)$ is determined by these. □

Proposition 19.6. Any k-form on $U \subset \mathbb{R}^n$ is given by $c_p = \sum_{|I|=k} f_I(p) \, dx_I$, where $f_I \colon U \to \mathbb{R}$ is smooth. Any k-form on M can be written this way on charts. In particular, $\omega(X_1, \ldots, X_k)(p) = \omega_p((X_1)_p, \ldots, (X_k)_p)$ for some well defined alternating k-tensor ω_p.

Proof. The same proof as for vector fields and 1-forms. □

§20 Day 20: Operations on Differential Forms (Mar. 24, 2026)

Last time, we defined a k-form on M to be a smotoh choice of alternating k-tensor

$$\omega_p \colon T_pM \times \cdots \times T_pM \to \mathbb{R}.$$

Formally, $\omega \colon \mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M) \to C^\infty(M)$ is an alternating $C^\infty(M)$-multilinear function. We may regard this a coordinate-wise construction,

$$\omega = \sum_{\substack{|I|=k \\ I \subset \{1,\ldots,n\}}} f_I \, dx_I,$$

where dx_I is 1 on $\partial_{i_1}, \ldots, \partial_{i_k}$ and zero on other sets of basis vectors. We have the following operations on differential forms,

- Wedge product,
- Exterior derivative,
- Pullback along functions,
- Integration over submanifolds.

For the wedge product (§7.22), we have that $\wedge \colon \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$. We start with the simplest case $k = \ell = 1$. Let $\alpha, \beta \in \Omega^1(M)$, where $\alpha_p \colon T_pM \to \mathbb{R}$ and $\beta \colon T_pM \to \mathbb{R}$. Is the wedge product defined like this, $\alpha \wedge \beta(v_1, v_2) = \alpha(v_1)\beta(v_2)$? No, as it is not alternating. Perhaps we could define it as

$$\alpha \wedge \beta(v_1, v_2) \coloneqq \alpha(v_1)\beta(v_2) - \alpha(v_2)\beta(v_1).$$

As an example, on $\mathbb{R}^2$, consider $\alpha = dx$ and $\beta = dy$. We have $\alpha \wedge \beta(v_1, v_2) = x_1y_2 - x_2y_1$ (where $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$), which is the determinant of the 2×2 matrix with rows v_1, v_2. On $\mathbb{R}^n$, let $\alpha = dx$ and $\beta = dy$ again; we have that

$$\alpha \wedge \beta(\partial_i, \partial_j) = 0$$

unless $\{i, j\} = \{1, 2\}$, so this specializes down to our case again, i.e., $\alpha \wedge \beta(\partial_1, \partial_2) = 1$ and $\alpha \wedge \beta(\partial_2, \partial_1) = -1$. In general, given a k-linear map $\varphi \colon V \times \cdots \times V \to \mathbb{R}$, we can product an alternating map

$$\mathrm{Sk}(\varphi)(v_1, \ldots, v_k) = \sum_{\sigma \in S_k} \mathrm{sign}(\sigma)\varphi(v_{\sigma(1)}, \ldots, v_{\sigma(k)}).$$

In this manner, define: for $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$, we have

$$\alpha \wedge \beta(v_1, \ldots, v_{k+\ell}) = \frac{1}{k!\ell!} \sum_{\sigma \in S_{k+\ell}} \mathrm{sign}(\sigma)\alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)})\beta(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+\ell)}),$$

i.e., we may "sum over all shuffles". In particular, $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ for $i_1 < \cdots < i_k$; why is this well-defined? We have the properties of the wedge product,

(i) Bilinearity: $(f\alpha_1 + \alpha_2) \wedge \beta = f\alpha_1 \wedge \beta + \alpha_@ \wedge \beta$,

(ii) Graded commutativity (§7.24): $\beta \wedge \alpha = (-1)^{k\ell}\alpha \wedge \beta$, where $\beta \in \Omega^\ell$ and $\alpha \in \Omega^k$. We may note that kl is the sign of the permutation sending $1, \ldots, k+\ell$ to $k+1, \ldots, k+\ell, 1, \ldots, k$.

(iii) Associativity (also §7.24): $(\alpha_1 \wedge \alpha_2) \wedge \alpha_3 = \alpha_1 \wedge (\alpha_2 \wedge \alpha_3)$.

Corollary 20.1 (§7.21)**.** Every form is locally a sum of forms of the form $f_0 df_1 \wedge \cdots \wedge df_k$, where $f_0, \ldots, f_k$ are smooth functions.

Proof sketch. $f_0\, dx_1 \wedge \cdots \wedge dx_k$ in local cordinates and $dx_i = d(\vec{x} \mapsto x_i)$. □

However, coordinate functions don't extend to the entire of M, so we may pick a compact $K \subset U$ (where U is the domain of a coordinate chart) with $k \in V$ open and $V \subset K$. Pick $\varphi\colon M \to \mathbb{R}$ to be supported on K and equal to 1 on V; then we may use functions $f_i = \varphi x_i$, which extend over M, and $\omega = \sum_{|I|=k, I \subset [n]} \omega_I df_{i_1} \wedge \cdots \wedge df_{i_k}$.[29]

We now discuss exterior derivatives. Let $\Omega^k \to \Omega^{k+1}$ be a generalization of "$f \mapsto df$".

Definition 20.2 (§7.25)**.** There's a unique collection of linear maps $\Omega^k(M) \to \Omega^{k+1}(M)$ for each k, such that

(i) For $f \in \Omega^0(M) = C^\infty(M)$, we have $df \in \Omega^1$ is the differential,

(ii) $d(df) = 0$ for all $f \in C^\infty(M)$,

(iii) The graded product rule holds $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge d\beta$.

(iv) $d(d\alpha) = 0$ for all $\alpha \in \Omega^k(M)$.

Proof outline. Assume existence, and use this to prove uniqueness, which we will derive a formula for. □

Lemma 20.3. d is local, i.e., $(d\alpha)|_U$ only depends on $\alpha|_U$, i.e., this is equivalent to, if $\alpha|_U = 0$, then so is $d\alpha|_U$.

Proof. Define f to be a function that is supported on U and equal to 1 near p, so $f\alpha = 0$. By (i, ii, iii), $0 = d(f\alpha) = df \wedge \alpha + f\, d\alpha$. Since df is 0 at p and f is 1 at p, we have $(d\alpha)|_p = 0$. We can do this for every $p \in U$. □

It's enough to see what happens on charts, $\alpha = \sum f_0\, df_1 \wedge \cdots \wedge df_k$. By (i, ii, iii), we have

$$d(f_0\, df_1 \wedge \cdots \wedge df_k) = df_0 \wedge \cdots \wedge df_k.$$

This gives us the formula in coordinates,

$$d(f\, dx_I) = df \wedge dx_I.$$

This guarantees uniqueness and the forms property. After this, we may check existence (the formula above is consistent over different charts). This was left as an exercise.

Definition 20.4. A k-form $\omega \in \Omega^k(M)$ is closed if $d\omega = 0$; it is exact if $\omega = d\alpha$ (for some α).

For some examples,

(i) $\omega \in \Omega^m(M)$ implies $d\omega = 0$. There are m nonzero $(m+1)$-forms on M.

(ii) $d\theta \in \Omega^1(S^1)$ is closed but not exact (on the other hand, every exact form is closed by (iv)).

[29] i've ran out of footnotes to write and fucks to give, $[n]$ is combinatorialists notation for 1 to n.

Definition 20.5. The *de Rham cohomology* of M is $H^k(M)$, defined as the quotient of closed forms by exact forms. This is a vector space.

Fact 20.6. $H^k(M)$ is finite dimensional if M is compact. $H^k(M)$ only depends on the topology of M (not diffeomorphism type).

We now define pullbacks of forms.

Definition 20.7. Given $F\colon M \to N$, $\alpha \in \Omega^k(M)$, we define the pullback by F to be $F^*\alpha(v_1, \dots, v_k) = \alpha(F_*v_1, \dots, F_*v_k)$.

The pullback admits the following properties (§7.29),

(i) The pullback is actually a differential form (it varies smoothly).

(ii) It preserves $\wedge$: $F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^\beta$,

(iii) It preserves d: $F^*(d\alpha) = dF^*\alpha$.

Proof. First, we check that this holds pointwise. Indeed,

$$F^*(\alpha \wedge \beta)(v_1, \dots, v_k) = (\alpha \wedge \beta)(F_*v_1, \dots, F_*v_k) = \sum \alpha(F_*v_{\sigma(1)}, \dots, F_*v\sigma(k))\beta(\dots),$$

which is precisely $F^* \wedge F^*\beta$. For the second step, observe that $d, \wedge, F^*$ are all $\mathbb{R}$-linear maps, and that they satisfy local properties, so we only have to show this for $f_0\, df_1 \wedge \cdots \wedge df_k$. From the beginning, we get that

$$F^*(df_0\, df_1 \wedge \cdots \wedge df_k) = F^*f_0 F^*df_1 \wedge \cdots \wedge F^*df_k,$$

which we've already seen that pullbacks of 1-forms are forms, so we have (i). We also know $F^*(df) = dF^*f$, so we get (iii). $\square$

In particular, we know that, for $F\colon \mathbb{R}^k \to \mathbb{R}^k$, every k-form is $\omega = f\, dx_1 \wedge \cdots \wedge dx_k$ for some f. We also have $F^*\omega(\partial_1, \dots, \partial_k) = \omega(F_*\partial_1, \dots, F_*\partial_k) = \omega(\partial_1 F, \dots, \partial_k F)$, which is precisely $\det(DF)\omega(\partial_1, \dots, \partial_k)$, whence $F^*\omega|_p = \det(DF_p)\omega|_{F(p)}$.

We now work through chapter 8 of the book. For a k-form on $\mathbb{R}^k$, every k-form is $\omega = f\, dx_1 \wedge \cdots \wedge dx_k$ for some $f \in C^\infty(\mathbb{R}^k)$. If f has compact support in U, then we can define $\int_U \omega = \int_{\mathbb{R}^k} f(x_1, \dots, x_k)\, dx_1 \dots dx_k$. From last time, we see that

$$\int_{F(u)} \omega = \pm \int_U F^*\omega$$

by change of variables, where $F\colon U \to F(U)$ is a diffeomorphism. For k-forms on a k-manifold, suppose $\omega \in \Omega^k(M)$ (where $\dim M = k$) has compact support; then we can find charts $U_1, \dots, U_r$ such that ω is supported on $U_1 \cup \cdots \cup U_r$, and we can write

$$\omega = \rho_1\omega + \cdots + \rho_r\omega,$$

where $\rho_i\colon M \to \mathbb{R}$, where the ρ_i sum to 1 on $\operatorname{supp}(\omega)$, and each ρ_i vanish outside of U_i. This is a partition of unity. By definition, we get

$$\int_M \omega = \sum_{i=1}^r \int_{\varphi_i(U_i)} (\varphi_i^{-1})^*(\rho_i\omega),$$

where we choose the φ_i's to have compatible orientations, i.e., $\det D(\varphi_i \circ \varphi_j^{-1}) > 0$ for all i, j.